



Contents lists available at ScienceDirect

Journal of International Money and Finance

journal homepage: www.elsevier.com/locate/jimf

Evolution of the exchange rate pass-through into prices in Peru: An empirical application using TVP-VAR-SV models [☆]

Gabriel Rodriguez ^{a,*}, Paul Castillo B. ^{b,a}, Roberto Calero ^a,
Rodrigo Salcedo Cisneros ^a, Miguel Ataurima Arellano ^{c,a}

^a Pontificia Universidad Católica del Perú, Peru

^b Central Reserve Bank of Peru, Peru

^c CAF-Development Bank of Latin America, Venezuela

ARTICLE INFO

JEL classification:

C11
C32
E31
F31

Keywords:

Exchange rate pass-through into prices
Vector autoregressive model with time-varying parameters
Stochastic volatility
Bayesian estimation and comparison of models
Deviance information criterion
Marginal likelihood
Peruvian economy

ABSTRACT

This study examines the evolution of exchange rate pass-through (ERPT) into import, producer, and consumer prices in Peru from 1995Q2 to 2022Q4 using time-varying parameter and stochastic volatility VAR models. Findings reveal a resurgence of ERPTs into import and producer prices since 2009, particularly during the period of a strong US dollar following the 2013 taper tantrum and from 2020 to 2022. Increased uncertainty surrounding the exchange rate and future macroeconomic policies, triggered by the political uncertainty following the 2021 general elections, may have contributed to this trend. Short-term ERPT exceeds long-term ERPT, which might reflect prevalent price dollarization in Peru's import and producer prices. Consumer ERPT remained stable at around 10% until 2009, then increased to 15%, indicating lower levels of price dollarization. This paper sheds light on ERPT dynamics in Peru, carrying implications for policymakers in emerging economies.

1. Introduction

Emerging market economies (EMEs) have made remarkable strides in implementing monetary policies that have led to lower and more stable inflation, especially in those that have adopted inflation targeting (IT) regimes and floating exchange rates.¹ These improvements have allowed them to weather various shocks successfully, including the Global Financial Crisis (GFC), the volatility in capital flows generated by advanced economies' non-conventional monetary policies, the reversal of these policies, and the COVID-19

[☆] This document is drawn from Roberto Calero and Rodrigo Salcedo Cisneros' Master's Thesis in Economics at the Graduate School of Pontificia Universidad Católica del Perú (PUCP). It represents a very substantial revised version of Calero et al. (2022). We are grateful for the valuable comments by Marco Vega (Central Reserve Bank of Peru and PUCP), Alvaro Jiménez (Fiscal Council of Peru), the Editor of the Journal, Professor Menzie D. Chinn and two anonymous Referees, all of whom have contributed to improving the document. Any remaining errors are our responsibility.

* Corresponding author at: Department of Economics, Pontificia Universidad Católica del Perú, 1801 Universitaria Avenue, Lima 32, Peru.

E-mail addresses: gabriel.rodriguez@pucp.edu.pe (G. Rodriguez), paul.castillo@bcrp.gob.pe (P. Castillo B.), roberto.calero@pucp.edu.pe (R. Calero), rodrigo.salcedo@pucp.edu.pe (R. Salcedo Cisneros), miguel.ataurima@pucp.edu.pe (M. Ataurima Arellano).

¹ In Latin America, Brazil, Colombia, Chile, Peru, Mexico and Paraguay implement their monetary policy using an inflation targeting (IT) regime with a floating exchange rate regime.

<https://doi.org/10.1016/j.jimonfin.2024.103023>

Available online 7 February 2024

0261-5606/© 2024 Elsevier Ltd. All rights reserved.

pandemic. In particular, emerging market economies in Latin America have kept inflation until 2020 low and stable while avoiding past financial and balance of payment crises.

One significant factor behind these achievements has been the better anchoring of inflation expectations and increased exchange rate flexibility, leading to a substantial reduction in the exchange rate pass-through (ERPT) into consumer prices. For instance, Maertens Odría et al. (2012) and Winkelried (2014) find evidence of lower ERPT after the adoption of a fully fledged inflation targeting regime in Peru. Lower ERPT of consumer prices allows greater flexibility of the real exchange rate in response to external shocks, facilitating adjustments to the economy without compromising the central bank's ability to control domestic inflation. However, ERPT can vary and be different among alternative price indexes, such as consumer prices, producer prices, and import prices, depending on various factors, including competition in the final and intermediate goods markets and the relative volatility of local firms' costs and the exchange rate.

The significance of ERPT for the design and implementation of monetary policy has been emphasized by researchers such as Maertens Odría et al. (2012), Winkelried (2014) and Sansone and Justel (2016). Moreover, both the level and velocity of ERPT can influence the transmission mechanism of monetary policy, as well as inflation control and forecasting, consequently, it is crucial for the monetary authority correctly estimate the impact of exchange rate fluctuations on inflation dynamics. However, assuming a time-constant ERPT could potentially bias its estimation and undermine the effectiveness of monetary policy.

To overcome the limitation of assuming a time-constant ERPT, this study estimates the evolution of ERPT into three types of prices in Peru from 1995Q2-2022Q2 using a time-varying stochastic volatility vector autoregression (TVP-VAR-SV) approach proposed by Chan and Eisenstat (2018). This methodology allows the parameters and stochastic volatility to vary over time, enabling a more accurate estimation of short-term and long-term ERPT and their dynamics across consumer, producer, and import prices. By capturing simultaneous, non-linear, and time-varying relationships between variables and the heteroscedasticity of shocks, this approach considers the effects of structural reforms, economic policy shifts, and external shocks experienced by Peru over the period of analysis.² It also extends, in a more efficient way, the estimation of ERPT's evolution over time conducted by Winkelried (2014) and generalizes the non-linear multivariate treatment, the analysis in sub-samples or regimes, and the estimation of long-term changes without sub-sampling. This flexible approach improves our understanding of the impact of exchange rate fluctuations on inflation dynamics and it can better inform the design of a more effective monetary policy.

Additionally, we incorporate into our analysis the COVID-19 period and the recent period of higher inflation, enhancing the value of our findings. Notably, we utilize a large data sample that includes the high-inflation period of 2022, which can significantly improve the accuracy of our estimation of the time evolution of ERPT. To comprehensively capture the effects of both domestic and foreign monetary policy and the relevance of terms of trade, our baseline model includes additional variables alongside the quarterly changes in import, producer, and consumer prices. Specifically, we incorporate the Federal Funds rate and the policy rate of the Peruvian Central Bank. These additional variables, along with the export price index, the growth rate of the nominal exchange rate, and Peruvian GDP, provide a comprehensive framework for analyzing the ERPT.

Peru provides an interesting case study for analyzing the time-varying evolution of the ERPT, as it shares many common characteristics with other economies in Latin America. Firstly, it is a commodity-exporting open economy, like Chile, Brazil and Colombia. Secondly, it follows an inflation targeting regime for monetary policy, like Brazil, Colombia, Chile, Mexico and Paraguay. Thirdly, it is also an economy with dollarization, a feature present in other economies in the region, such as Paraguay, Uruguay, Ecuador and many Central American countries. These common features have an impact on the ERPT. On one hand, an inflation targeting regime can contribute to reduce the ERPT by decreasing inflation and inflation expectations' uncertainty, which in turn makes local prices less sensitive to supply shocks. See Maertens Odría et al. (2012) for empirical evidence about the link between the adoption of inflation targeting and the evolution of the ERPT for Peru. Additionally, the fraction of domestic prices invoiced in dollars, known as price dollarization, can also have a direct impact on the ERPT when prices are sticky in the short run. These common features make Peru a natural case study for quantifying the time-varying evolution of short- and long-term ERPT.

Our study utilizes two Bayesian selection criteria, the Marginal Log-Likelihood and the Deviation Information Criterion, to identify models that best fit the data. Our results suggest that models that allow some parameters to change over time, particularly the stochastic volatility (SV), provide a better fit to the data.³ The estimation of time-varying ERPT yields three main findings. Firstly, the ERPTs into import and producer prices are significantly higher than that of consumer prices and exhibit greater variation over time. They show a reduction starting in 2004 until 2009, just after Peru adopted its fully fledged inflation targeting regime in 2002. However, after the GFC, both ERPTs show a marked resurgence until the end of the sample. It is worth to mention that previous papers that study ERPT in Peru have found a reduction of the ERPT up to 2011 which is consistent with our results; see Winkelried's study (2014) who estimated the evolution of ERPT using a VAR model with rolling windows, covering data until 2011.

Secondly, the median short-term ERPT into import and producer prices is higher than the long-term ERPT. For import prices, the short-term ERPT is approximately 1, while the long-term ERPT is approximately 0.8. For producer prices, the short-term ERPT is around 0.6, while the long-term ERPT is approximately 0.5. Finally, the ERPT into consumer prices is around 0.1 for both the short-term and long-term, suggesting that the transmission of exchange rate movements into prices varies depending on the time horizon and on the type of price index. The low and stable ERPT into consumer prices is consistent with the low inflation and stable

² During this period, Peru experimented a change in monetary policy framework, from the control of monetary aggregates to a fully-fledged inflation targeting regime in 2002, changes in tariffs, and monetary and fiscal policy instruments.

³ Other papers have focused on estimating the levels of ERPTs and how these levels change in response to currency appreciation or depreciation, such as the works of Pérez-Forero and Vega (2015) and Cueva (2018), or within specific sub-samples or regimes, as examined by Maertens Odría et al. (2012).

inflation observed in the Peruvian economy since the adoption of IT regime -see Maertens Odría et al. (2012), Winkelried (2014), BIS (2019); and also with low levels of price dollarization for consumer prices, see Castellares and Toma (2020).

The observed dynamic might be attributed to varying degrees of price dollarization among import, producer, and consumer prices. Consumer prices exhibit minimal price dollarization, less than 5%, whereas producer prices can be influenced up to 40% by price dollarization, and import prices can reach approximately 60% of dollarization. When local consumer prices are denominated in foreign currency, exchange rate fluctuations immediately impact these prices due to their short-run inflexibility in dollars. However, since the demand for these goods and services depends on local factors, the resulting increase in relative prices due to exchange rate changes leads to higher costs for consumers, resulting in reduced demand and ultimately a decline in prices denominated in US dollars, and as result the long-run ERPT gradually falls. This finding carries significant implications for policymakers, indicating that inflation forecasts in dollarized economies should account for the higher short-term ERPT compared to the long-term perspective. More importantly, our findings provide a rationale for the more active use of foreign exchange intervention in economies with significant dollarization, as observed in countries like Peru and Costa Rica in Latin America. Due to the stickiness of dollar prices, exchange rate fluctuations temporarily disrupt relative prices, moving them away from their long-term equilibrium. This effect has adverse consequences for resource allocation efficiency. Therefore, central banks that engage in foreign exchange market interventions to mitigate volatility can help reduce the divergence in relative prices resulting from abrupt, short-term exchange rate movements, minimizing the welfare costs associated with nominal rigidities. This novel finding sets our work apart from previous literature on ERPT in Peru, which primarily concentrates on quantifying ERPT and its trends but doesn't delve into the analysis of disparities between short and long-term ERPT for consumer, producer, and imported prices, or their association with varying degrees of price dollarization across the supply chain. Additionally, previous studies often omit discussions about the potential policy implications arising from these linkages.

Our analysis has also revealed that the Peruvian economy experienced a significant increase in the ERPT during the period of high uncertainty from 2020 to 2022. Specifically, we observed an increase in the short-term ERPT into import and producer prices from 1.2 to 1.6 and from 0.65 to 0.85, respectively. However, it is important to note that the long-term ERPT increased less, from 1.0 to 1.2 for import prices and 0.65 to 0.75 for producer prices. This pattern suggests that more firms may have started to set their prices in dollars since 2020, possibly due to increased uncertainty about both the exchange rate and future macroeconomic policies, including the potential for regime change after the 2021 presidential election. These findings have important implications for policymakers and businesses seeking to anticipate and manage inflationary pressures in the Peruvian economy. A higher ERPT can complicate the trade-offs faced by central banks when supply shocks impact the economy. A temporary yet persistent exchange rate increase can yield effects akin to those resulting from supply shocks, such as inflation spikes and amplified costs for imported goods like oil, which are priced in domestic currency. With an elevated ERPT, the inflationary impact can be more pronounced, potentially unsettling inflation expectations. This, in turn, might prompt the central bank to raise its policy rate as a preemptive measure against second-round inflationary effects triggered by these altered expectations. Failure to respond promptly and decisively could result in inflation persistently deviating from the target range for an extended period, leading to more pronounced adverse economic consequences. In that sense, higher ERPT combined with the sharp upturn in oil and food prices during these years, may offer an explanation for the rapid surge in import and producer prices witnessed in Peru throughout 2021 and 2022; see Cornejo et al. (2022) for a decomposition of the contribution of external shocks to domestic inflation in Peru.

All the findings are robust to considering four robustness exercises: (i) estimation of baseline model with Lenza and Primiceri (2022) Covid-19 pandemic adjustment, (ii) changing the Federal Reserve Funds Rate for the shadow federal funds rate proposed by Wu and Xia (2016); (iii) changing the export prices inflation rate for the Global Price of Metal Index growth rate; and (iv) identification via sign restrictions scheme.

These findings are in line with the recent literature that uses the TVP-VAR-SV approach and postulates the resurgence of ERPT after the GFC -see Jooste and Jhaveri (2014), Alexius and Holmberg (2017), and Dahem et al. (2017). Recent studies like Dahem et al. (2017) and Baxa and Šestořád (2019), based on Primiceri (2005), Cogley and Sargent (2005), and Nakajima (2011), estimate the ERPT for several economies using TVP-VAR-SV models, showing that its behavior is not constant either over time or in sub-samples; and that the IT adoption as well as shocks like the GFC have implications for decline and dynamics of ERPT; see Forbes et al. (2018) and Jašová et al. (2019), among others. These papers show that the ERPT increases during and after the GFC, and in episodes of depreciation of the domestic currency against the US dollar. See Shioji (2014, 2015) and Alexius and Holmberg (2017) for advanced economies and Jooste and Jhaveri (2014) and Dahem et al. (2017) for EMEs. The use of TVP-VAR-SV models for analysis of ERPTs in EMEs is still incipient except for Jooste and Jhaveri (2014) and Dahem et al. (2017). In this light, this paper adheres to this approach in order to contribute to estimate the ERPT for Peru.

The remainder of the document is organized as follows. Section 2 reviews the relevant empirical literature. Section 3 describes the methodology. Sections 4 and 5 present the empirical results and the robustness analysis, respectively. Finally, Section 6 shows the conclusions.

2. Literature review

Empirically, the ERPT is estimated via several approaches, mainly: (i) Dynamic Stochastic General Equilibrium (DSGE) models; (ii) cointegration models; (iii) single-equation models with constant parameters and time-varying parameters; and (iv) constant

Vector Autoregressive (VAR) models and time-varying parameters (TVP) VAR models with/without stochastic volatility (SV). This paper is located in this last group of methodologies.⁴

The literature using VAR models for ERPT estimation is extensive and varied regarding the specification and identification scheme for structural shocks; see Aron et al. (2014) and Tunç (2017) for a comprehensive review. Studies using linear VAR models with recursive identification include the seminal work by McCarthy (2007) for a set of industrialized economies, Ca'Zorzi et al. (2007) for Mexico and Chile, Miller (2003) and Winkelried (2014) for Peru, and Helmy et al. (2018) for Egypt.⁵

All studies using VAR models (irrespective of the method for identifying structural shocks) assess mainly the effect of IT adoption on the ERPT; and conclude that the latter: (i) declined considerably since IT adoption; (ii) is incomplete (less than the unity) and time-varying; and (iii) depends on the exchange rate shock. Moreover, exchange rate variations show a steadily declining impact on pricing along a distribution chain (PDC); and this impact is smaller in low-inflation economies.

At the same time, few studies use non-linear VAR models to analyze and estimate the ERPT; and most of them use recursive identification. These studies focus on the asymmetry and influence of the state of the economy on ERPT estimation. For Peru, Winkelried (2003) uses a smooth-transition VAR model (ST-VAR); Maertens Odría et al. (2012) use an ST-VAR model with two regimes (pre- and post-IT); and Pérez-Forero and Vega (2015) use a non-linear VAR model estimated with Bayesian techniques. The results indicate that the ERPT: (i) is non-linear and (ii) declined since IT adoption; and (iii) exchange rate depreciation shocks have a higher impact than appreciation shocks.⁶

Studies using non-linear VAR models for other Latin American economies include Rincón-Castro and Rodríguez-Niño (2018), who use a logistic Bayesian ST-VAR model for Colombia. For Mexico, Aleem and Lahiani (2014) use a threshold VAR (T-VAR) model with three regimes; and Donayre and Panovska (2016) use a Bayesian T-VAR model. For Brazil, Balcilar et al. (2019) use an ST-VAR model. In general, the results indicate that the ERPT: (i) is non-linear; (ii) declined since IT adoption; (iii) exchange rate depreciation shocks have a higher impact than appreciation shocks; (iv) depends on the state of the economy and the type of shock; (v) is greater when the size of the shock exceeds the threshold; (vi) is complete for import prices; and (vii) prices react significantly to exchange rate shocks. We also underscore the studies by Hernández and Leblebicioğlu (2012) and Khemiri and Ben Ali (2013), who employ Markov-switching models to analyze the ERPT in the US and Tunisia, respectively.

However, linear or non-linear multivariate time-series (where the changes in the parameters are captured via estimation of the model by sub-samples or regimes) methodologies do not consider all structural/policy changes and external shocks affecting the economy, nor the evolving nature of the exchange rate. In this context, there are few studies using TVP-VAR-SV models and variants to estimate the ERPT. The studies by Shioji (2012, 2014, 2015) and Çatık et al. (2016) use the TVP-VAR model proposed by Cogley and Sargent (2001). For their part, Clark and Terry (2010), Arratibel and Michaelis (2014), Jooste and Jhaveri (2014), Alexius and Holmberg (2017), Dahem et al. (2017), and Baxa and Šestorád (2019) use TVP-VAR-SV models based on Primiceri (2005), Cogley and Sargent (2005), and Nakajima (2011) while Moussa (2016) uses a factor-augmented VAR model with time-varying parameters (TVP-FAVAR).

Shioji (2012) analyzes the ERPT into import prices and a set of export and domestic prices (at the aggregate and disaggregate level) over time for Japan, finding that the ERPT into import prices is higher than into aggregate consumer prices; and that it declines considerably since the end of the 1990s. Additionally, the ERPT into aggregate consumer prices shows a fall since the 1980s; and the ERPT into domestic intermediate good prices drops significantly since 1980. Additionally, Shioji (2014) estimates the ERPT and the pass-through of import prices into a set of domestic prices in Japan. The author identifies a fall in the ERPT in 1980-1995 and a resurgence since 2012, except for the ERPT into service consumer prices. ERPT evolution is largely explained by Japan's cost structure changes over time.

In the same line, Shioji (2015) revisits the ERPT into consumer prices in Japan in a context of high yen depreciation and close-to-zero interest rates; and identifies a significant ERPT resurgence in recent years for regular household consumption goods (e.g., gasoline). For their part, Çatık et al. (2016) analyze the evolution of the ERPT into import, producer, and consumer prices in Turkey; and find evidence of an evolving behavior over time. They also find that the ERPT into import prices is low and non-significant; and that the ERPTs into producer and consumer prices show a declining trend throughout the period of analysis, explained by IT adoption. However, they highlight that the ERPTs into producer and consumer prices peaked during the 1994 financial crisis and increased slightly during the GFC.

For their part, Clark and Terry (2010) analyze the pass-through from energy prices into core inflation in the US; and identify a significant pass-through fall over 1974Q1-1985Q1, explained by declining energy consumption. Additionally, Arratibel and Michaelis (2014) study the reaction of output and prices to interest rate and exchange rate shocks in Poland, finding that the ERPT has fallen significantly over time; and that the ERPT into import prices is higher than into producer or consumer prices, thereby supporting the evidence of ERPT reduction along the distribution chain.

Jooste and Jhaveri (2014) estimate the ERPT and identify their main determinants for South Africa, finding that the ERPT changes over time and depends on the state of the economy. The ERPT peaked in 2002-2003, in a context of considerable exchange

⁴ Literature about the first three approaches include: Kim (1990), Darvas (2001), Albuquerque and Portugal (2005), Sekine (2006), De Bandt et al. (2008), Takhtamanova (2010), De Souza et al. (2013), Ben Ali and Tarek (2014), Buyandelger (2015), Hara et al. (2015), Bada et al. (2016), Jiménez-Rodríguez and Morales-Zumaquero (2016), López-Villavicencio and Mignon (2016), Ponomarev et al. (2016), Razafindrabe (2016), Liu and Chen (2017), Forbes et al. (2018), Palleja (2019), Pan (2018), Patra et al. (2018), Jašová et al. (2019), Marodin and Portugal (2019), and Gayaker et al. (2021).

⁵ Studies using non-recursive identification of structural shocks include Shambaugh (2008), An and Wang (2012), Karagöz et al. (2016), Borensztein and Queijo (2016), Comunale and Kunovac (2017), Corbo and Di Casola (2018), and Tunç and Kilınç (2018).

⁶ Cueva (2018) also analyzes non-linearity and asymmetry of the ERPT in Peru and Mexico, although employing the semiparametric local projection technique.

rate volatility and inflation; but began to decline since 2004. Additionally, they indicate that the average ERPT does not change significantly between the pre- and post-IT periods. For their part, Alexius and Holmberg (2017) analyze the effect over time of the ERPT and foreign prices on domestic inflation for a set of large (Australia, Japan, UK and US) and small (Canada, New Zealand, Sweden and Switzerland) economies with a floating exchange rate regime and low inflation, finding that a unit exchange rate shock has a greater effect on inflation in small economies; and that the pass-through of external prices into domestic prices is much larger and instantaneous than the ERPT. Moreover, they find that the ERPT into domestic prices: (i) peaks during the GFC and (ii) resurges post-GFC (the 2014-2015 low-inflation period) in Australia and the US, as inflation deviated from target.

Dahem et al. (2017) study the evolution over time of the effect of monetary and exchange rate shocks on prices in Tunisia, finding a time-varying ERPT; and that the ERPT into government-controlled prices (food and fuel) surges after the 2010-2011 political developments in that country. For their part, Baxa and Šestořád (2019) estimate the ERPT and the pass-through into output growth for the Czech Republic, in a context of depreciation and low inflation, finding that the ERPT into consumer prices is incomplete but instantaneous; and that it has declined significantly since the 2000s. However, they also find that depreciation of the Czech Koruna against the Euro pushed prices up at end-2013.

Finally, Moussa (2016) uses a TVP-FAVAR model to analyze ERPT evolution in Japan, finding that it differs across prices in magnitude and time variation; and that it declines along the distribution chain. Additionally, the study finds that the ERPT into import and domestic prices (at the aggregate and disaggregate level) fell since 1985 until the end of the 2000s (except for the Asian crisis) and resurged thereafter through the end of the sample.

The review of the literature shows that use of the TVP-VAR-SV approach for ERPT estimation is a work in progress. However, given its specification features, this approach is usually more appropriate for ERPT estimation than studies using rolling-window VAR models or single-equation models with time-varying parameters. At the same time, most studies employing TVP-VAR-SV models and their variants focus on developed countries, and their use for EMEs is still incipient except for Jooste and Jhaveri (2014) and Dahem et al. (2017). In this light, this paper adheres to the TVP-VAR-SV approach to estimate the ERPT for Peru following the methodology proposed by Chan and Eisenstat (2018), which we describe below.

3. Methodology

Following the methodology proposed by Chan and Eisenstat (2018), we use a set of TVP-VAR-SV models to estimate the time-varying ERPT.

3.1. General model: TVP-VAR-SV

Following Chan and Eisenstat (2018), a TVP-VAR-SV model has the following specification:

$$\mathbf{B}_{0,t}\mathbf{y}_t = \boldsymbol{\mu}_t + \mathbf{B}_{1,t}\mathbf{y}_{t-1} + \dots + \mathbf{B}_{p,t}\mathbf{y}_{t-p} + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_t), \quad (1)$$

where $\boldsymbol{\mu}_t$ is an $n \times 1$ vector of time-varying intercepts, $\mathbf{B}_{1,t} \dots \mathbf{B}_{p,t}$ are the $n \times n$ matrices of coefficients associated with the vector of lagged endogenous variables, $\mathbf{B}_{0,t}$ is the $n \times n$ lower triangular matrix of contemporaneous effects with diagonal unit values, and $\boldsymbol{\Sigma}_t = \text{diag}(\exp(h_{1,t}), \dots, \exp(h_{n,t}))$. The movement law for the logs of all variables $\mathbf{h}_t = (h_{1,t}, \dots, h_{n,t})'$ is specified as an independent random walk:

$$\mathbf{h}_t = \mathbf{h}_{t-1} + \boldsymbol{\zeta}_t, \quad \boldsymbol{\zeta}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_h), \quad (2)$$

where the initial conditions $\mathbf{h}_0$ are also parameters to be estimated.

As the system in (1) is in structural form and the variance matrix $\boldsymbol{\Sigma}_t$ is diagonal, the estimation can be carried out recursively. For this purpose, we rewrite the model. We consider the $k_\beta \times 1$ vector of intercepts and coefficients associated with the lagged observations $\boldsymbol{\beta}_t = \text{vec}((\boldsymbol{\mu}_t, \mathbf{B}_{1,t}, \dots, \mathbf{B}_{p,t})')$. The second $k_\gamma \times 1$ vector, containing the time-varying coefficients that characterize contemporaneous relationships between variables, is denoted by $\boldsymbol{\gamma}_t$. It should be noted that $k_\beta = n(np + 1)$ and $k_\gamma = n(n - 1)/2$. Therefore, equation (1) can be rewritten as:

$$\mathbf{y}_t = \tilde{\mathbf{X}}_t \boldsymbol{\beta}_t + \mathbf{W}_t \boldsymbol{\gamma}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_t),$$

where $\tilde{\mathbf{X}}_t = \mathbf{I}_n \otimes (\mathbf{1}, \mathbf{y}'_{t-1}, \dots, \mathbf{y}'_{t-p})$ and $\mathbf{W}_t$ is an $n \times k_\gamma$ matrix containing the appropriate elements of $-\mathbf{y}_t$.⁷ If $\mathbf{X}_t = (\tilde{\mathbf{X}}_t, \mathbf{W}_t)$, we can simplify the above model to obtain a generic space-state representation:

$$\mathbf{y}_t = \mathbf{X}_t \boldsymbol{\theta}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_t), \quad (3)$$

⁷ For example, when $n = 3$, $\mathbf{W}_t$ has the form:

$$\mathbf{W}_t = \begin{bmatrix} 0 & 0 & 0 \\ -y_{1,t} & 0 & 0 \\ 0 & -y_{1,t} & -y_{2,t} \end{bmatrix}$$

where y_{it} is the i th element of $\mathbf{y}_t$ for $i = 1, 2$.

where $\theta_t = (\beta'_t, \gamma'_t)'$ has a $k_\theta = k_\beta + k_\gamma$ dimension and the coefficients have a random walk behavior:

$$\theta_t = \theta_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \Sigma_\theta), \quad (4)$$

where the initial conditions θ_0 are also parameters to be estimated. The values of priors and hyperparameters are defined in Section 4.2.

3.2. Restricted models

Equation (1) defines (i) the general TVP-VAR-SV model. Based on the latter, as in Chan and Eisenstat (2018), we use restricted models, considering the parameters that we choose to keep constant: (ii) TVP-VAR: VAR with time-varying parameters and homoscedastic variance ($\mathbf{h}_t = \mathbf{h}_0$); (iii) TVP-VAR-R1-SV: VAR with constant parameters for the lagged variables and the intercepts ($\beta_t = \beta_0$) and SV; (iv) TVP-VAR-R2-SV: VAR with constant parameters for the contemporaneous relationships ($\gamma_t = \gamma_0$) and SV; (v) TVP-VAR-R3-SV: VAR with time-varying intercepts and SV; (vi) CVAR-SV: VAR with constant intercepts, parameters for the lagged variables ($\beta_t = \beta_0$), and parameters for the contemporaneous relationships ($\gamma_t = \gamma_0$) but with SV; and (vii) CVAR: VAR with constant parameters and homoscedastic variance.

3.3. Estimation algorithm: Gibbs sampling⁸

We estimate the posterior parameters using the Gibbs sampling method, which consists in dividing the parameters in blocks and estimating each one separately, conditional on updates in the other blocks. The draws are based on the precision sampling proposed by Chan and Jeliazkov (2009) and developed by Chan and Eisenstat (2018). The algorithm for estimating the (unrestricted) TVP-VAR-SV model is described as follows: (i) the draws are obtained from $(\theta|\mathbf{y}, \mathbf{h}, \Sigma_\theta, \Sigma_h, \theta_0, \mathbf{h}_0) \sim \mathcal{N}(\hat{\theta}, \mathbf{K}_\theta^{-1})$, where $\mathbf{K}_\theta = \mathbf{H}'_\theta \mathbf{S}_\theta^{-1} \mathbf{H}_\theta + \mathbf{X}' \Sigma^{-1} \mathbf{X}$ and the mean $\hat{\theta} = \mathbf{K}_\theta^{-1} (\mathbf{H}'_\theta \mathbf{S}_\theta^{-1} \mathbf{H}_\theta \alpha_\theta + \mathbf{X}' \Sigma^{-1} \mathbf{y})$, with $\alpha_\theta = \mathbf{H}_\theta^{-1} \tilde{\alpha}_\theta$. The matrices $\mathbf{H}_\theta$, $\mathbf{S}_\theta$, Σ and $\tilde{\alpha}_\theta$ are described in Appendix A of Chan and Eisenstat (2018); (ii) for the draws of $(\mathbf{h}|\mathbf{y}, \theta, \Sigma_\theta, \Sigma_h, \theta_0, \mathbf{h}_0)$, the auxiliary mixture sampler of Kim et al. (1998) is used; (iii) using the conditional distributions of the diagonal elements in Σ_θ , the draws are obtained from $(\sigma_{\theta_i}^2 | \mathbf{y}, \theta, \mathbf{h}, \theta_0, \mathbf{h}_0) \sim IG(v_{\theta_i} + \frac{T}{2}, S_{\theta_i} + \frac{1}{2} \sum_{t=1}^T (\theta_{it} - \theta_{i,t-1})^2)$ for $i = 1, \dots, k_\theta$ and the hyperparameters v_{θ_i} and S_{θ_i} are defined in Section 4.2; (iii) the draws are obtained from the diagonal elements in Σ_h with the form $(\sigma_{h_j}^2 | \mathbf{y}, \theta, \mathbf{h}, \theta_0, \mathbf{h}_0) \sim IG(v_{h_j} + \frac{T}{2}, S_{h_j} + \frac{1}{2} \sum_{t=1}^T (h_{jt} - h_{j,t-1})^2)$ for $j = 1, \dots, k_h$ and the hyperparameters v_{h_j} and S_{h_j} are defined in Section 4.2; (iv) the draws are obtained for the initial condition θ_0 from $(\theta_0 | \mathbf{y}, \theta, \mathbf{h}, \Sigma_\theta, \Sigma_h) \sim \mathcal{N}(\hat{\theta}_0, \mathbf{K}_{\theta_0}^{-1})$, where $\mathbf{K}_{\theta_0} = \mathbf{V}_\theta^{-1} + \Sigma_\theta^{-1}$ and $\hat{\theta}_0 = \mathbf{K}_{\theta_0}^{-1} (\mathbf{V}_\theta^{-1} \mathbf{a}_\theta + \Sigma_\theta^{-1} \theta_1)$ and values for $\mathbf{a}_\theta$ and $\mathbf{V}_\theta$ are given in Section 4.2; (v) the draws are obtained for the initial condition $\mathbf{h}_0$ from $(\mathbf{h}_0 | \mathbf{y}, \theta, \mathbf{h}, \Sigma_\theta, \Sigma_h) \sim \mathcal{N}(\hat{\mathbf{h}}_0, \mathbf{K}_{h_0}^{-1})$, where $\mathbf{K}_{h_0} = \mathbf{V}_h^{-1} + \Sigma_h^{-1}$ and $\hat{\mathbf{h}}_0 = \mathbf{K}_{h_0}^{-1} (\mathbf{V}_h^{-1} \mathbf{a}_h + \Sigma_h^{-1} \mathbf{h}_1)$ and values for $\mathbf{a}_h$ and $\mathbf{V}_h$ are given in Section 4.2; (vi) steps (i)-(v) are repeated N times. The estimation of the other (restricted) models follow similar stages to those previously described; see Appendix A of Chan and Eisenstat (2018) for further details.

3.4. Selection criteria

For comparing models and selecting the best fit, we use the log of the Marginal Likelihood calculated via the cross-entropy method ($\log \text{ML}_{CE}$) and the Deviance Information Criterion (DIC).

3.4.1. Marginal likelihood (ML_{CE})⁹

Chan and Eisenstat (2015) propose a better alternative for estimating the marginal likelihood using the cross-entropy method (ML_{CE}). This estimation is based on the importance sampling density $g(\theta_n)$:

$$\hat{p}_{IS}(\mathbf{y}) = \frac{1}{N} \sum_{n=1}^N \frac{p(\mathbf{y}|\theta_n) p(\theta_n)}{g(\theta_n)}, \quad (5)$$

where $\theta_1, \dots, \theta_N$ are the independent draws obtained from the importance sampling density. The $\hat{p}_{IS}$ estimator is consistent and unbiased irrespective of the value of $g(\theta_n)$, but is sensitive to the $g(\theta_n)$ variance. If the importance sampling is denoted by g^* , and using the posterior density to represent it, we infer that $\hat{p}_{IS}(\mathbf{y})$ is equivalent to $p(\mathbf{y})$. Therefore, the solution is choosing a g similar to g^* such that the variance of the estimator is minimized. We obtain g via the cross-entropy method, which is used to measure the distance between two densities.

Given the parametric family $\mathcal{F} = \{f(\theta, \mathbf{v})\}$ indexed by vector $\mathbf{v}$, we need to select the importance sampling $f(\theta, \mathbf{v}) \in \mathcal{F}$ that is closer to g^* . Therefore, it is necessary to choose the density $f(\theta, \mathbf{v}_{ce}^*) \in \mathcal{F}$ that minimizes the cross-entropy distance between the optimal density g^* and the chosen density $f(\theta, \mathbf{v})$ as follows:

⁸ Complete details about the algorithm for estimating the TVP-VAR-SV model and other restricted models can be found in Section 4 and Appendix A of Chan and Eisenstat (2018).

⁹ Complete details may be found in Section 4 and Appendix B of Chan and Eisenstat (2018).

$$\mathbf{v}_{ce}^* = \arg \min_{\{\mathbf{v}\}} \int g^*(\theta) \log g^*(\theta) d\theta - p(\mathbf{y})^{-1} \int p(\mathbf{y}|\theta) p(\theta) \log f(\theta, \mathbf{v}) d\theta,$$

$$\mathbf{v}_{ce}^* = \arg \max_{\{\mathbf{v}\}} \int p(\mathbf{y}|\theta) p(\theta) \log f(\theta, \mathbf{v}) d\theta,$$

whose estimator is:

$$\hat{\mathbf{v}}_{ce}^* = \arg \max_{\{\mathbf{v}\}} \frac{1}{R} \sum_{r=1}^R \log f(\theta_r, \mathbf{v}), \quad (6)$$

and we obtain the draws $\theta_1, \dots, \theta_R$ using the posterior density.¹⁰

3.4.2. Deviance information criterion (DIC)

The deviance information criterion (DIC) was proposed initially by Spiegelhalter et al. (2002). Based on Chan and Grant (2016), the deviance of the model's goodness of fit is defined as:

$$D(\theta) = -2 \log f(\mathbf{y}|\theta) + 2 \log h(\mathbf{y}), \quad (7)$$

where $f(\mathbf{y}|\theta)$ is the model's likelihood function and $h(\mathbf{y})$ is a function of the data. Additionally, we use a measure of model complexity via the effective number of parameters, defined as:

$$p_D = \overline{D(\theta)} - D(\tilde{\theta}), \quad (8)$$

where $\overline{D(\theta)} = -2E_\theta[\log f(\mathbf{y}|\theta)|\mathbf{y}] + 2 \log h(\mathbf{y})$ is the posterior mean deviance and $\tilde{\theta}$ is an estimate of θ (posterior mean or mode). Using these definitions, the DIC can be represented as the sum of the mean posterior deviation and the effective number of parameters; i.e., $DIC = \overline{D(\theta)} + p_D$. Assuming $h(\mathbf{y}) = 1$ and substituting the previous definitions, we obtain:

$$DIC = -4E_\theta[\log f(\mathbf{y}|\theta)|\mathbf{y}] + 2 \log f(\mathbf{y}|\tilde{\theta}), \quad (9)$$

where the estimate $\tilde{\theta}$ of θ is set as the posterior mode $\hat{\theta}$ and the first term of (9) can be estimated by averaging the log-integrated likelihoods $\log f(\mathbf{y}|\theta)$ over the posterior draws of θ . In order to approximate the posterior mode $\hat{\theta}$, we obtain the parameter set that yields the maximum value for $f(\mathbf{y}|\theta)f(\theta)$, where $f(\theta)$ is the prior density. Finally, the version used is the following:

$$DIC = -4E_\theta[\log f(\mathbf{y}|\theta)|\mathbf{y}] + 2 \log f(\mathbf{y}|\hat{\theta}).$$

There are other versions of this selection criterion based on complete-data likelihood or conditional likelihood.¹¹ However, in this study we used the DIC based on integrated likelihood, taking into consideration the results obtained by Chan and Grant (2016), since other DIC forms favor over-parameterized models and yield high standard errors.

4. Empirical results

4.1. Data

The variables in the vector of endogenous variables $\mathbf{y}_t$ are based on the PDC model proposed by McCarthy (2007), where the identification of three stages in domestic pricing (import, producer, and consumer prices; i.e. the first, second, and third stages, respectively) is instrumental in analyzing their reaction to financial, exchange rate, external (and domestic) demand shocks. In particular, $\mathbf{y}_t$ is composed of eight variables. The ones associated with inflation at each PDC stage are import, producer, and consumer price inflation (IPI, PPI and CPI, respectively). The variables that complete the model are the Federal Reserve funds rate (FED), the growth rate of the Export Price Index (XPI), and the growth rate of the gross domestic product (GDP), the domestic monetary policy interest rate (DMP), and the growth rate of the nominal exchange rate (NER).¹² With the exception of FED and DMP, the rest of the variables were seasonally adjusted using Census X-13 and expressed as quarterly growth rates.¹³

The selected period is limited by data availability. Information from the BCRP and the National Statistics Institute (INEI) databases is available for the 1995Q2-2022Q2 period, with a total of 109 observations for each variable.

¹⁰ Gelfand and Dey (1994), Chib (1995) and Chib and Jeliazkov (2001), among others, propose alternative methods for calculating the marginal likelihood. However, Frühwirth-Schnatter and Wagner (2008) show that using the conditional likelihood or the complete data likelihood obtained through the method suggested by Chib (1995) results in an incorrect choice of models. Moreover, Chan and Eisenstat (2015) use empirical results to show that the cross-entropy method is faster and more accurate than the three mentioned before.

¹¹ Spiegelhalter et al. (2002) provide the DIC expressions presented here; and Celeux et al. (2006) propose up to eight DIC versions, each with a different calculation method for likelihood according to the treatment of latent variables.

¹² The series employed are: Federal Reserve Fund Rate, Export Price Index (2007 = 100), Gross Domestic Product (millions of 2007 PEN), nominal exchange rate PEN/US Dollar, Domestic Policy Monetary (BCRP Reference Rate), Wholesale Import Price Index (2021 = 100), Wholesale Price Index (2021 = 100) and Consumer Price Index (2021 = 100).

¹³ With the exception of GDP, the rest of the variables used were constructed from monthly data. FED, DMP, IPI, PPI and CPI quarterly data was built from monthly data, the value of the last month of the quarter was taken to calculate the quarterly growth rates. Meanwhile, XPI and NER quarterly growth rates were obtained as quarterly averages of the monthly data for each quarter.

Fig. 1 shows the evolution of the variables in the baseline model, highlighting the declining trend of CPI since the beginning of the sample until around 2000, as a result of the macroeconomic stabilization process launched in the mid-1990s, and a slight increase in the last part of the sample as a consequence of the recent wave of inflation worldwide. It also shows that the behavior of nominal exchange rate variation is more volatile around domestic and external events of economic and political impact (i.e., the Russian-Brazilian and Asian crises at the end of the 1990s, the 2008 GFC, the Covid-19 Pandemic and Peruvian presidential election of 2021).

4.2. Priors

The priors of the initial conditions θ_0 and $\mathbf{h}_0$ are both Gaussian: $\theta_0 \sim \mathcal{N}(\mathbf{a}_\theta, \mathbf{V}_\theta)$ and $\mathbf{h}_0 \sim \mathcal{N}(\mathbf{a}_h, \mathbf{V}_h)$. We also assume that the error covariance matrices for the state equations are diagonal; i.e., $\Sigma_\theta = \text{diag}(\sigma_{\theta_1}^2, \dots, \sigma_{\theta_n}^2)$ and $\Sigma_h = \text{diag}(\sigma_{h_1}^2, \dots, \sigma_{h_n}^2)$. The elements of Σ_θ and Σ_h are independently distributed as $\sigma_{\theta_i}^2 \sim \text{IG}(v_{\theta_i}, S_{\theta_i})$, $i = 1, \dots, k_\theta$, $\sigma_{h_j}^2 \sim \text{IG}(v_{h_j}, S_{h_j})$, $j = 1, \dots, k_h$ where IG represents the Inverse Gamma distribution. The priors established for the hyperparameters are non-informative in all models. For the general TVP-VAR-SV model, we establish that $\mathbf{a}_\theta = \mathbf{0}$, $\mathbf{V}_\theta = 10 \times \mathbf{I}_{k_\theta}$, $\mathbf{a}_h = \mathbf{0}$, $\mathbf{V}_h = 10 \times \mathbf{I}_n$, and $v_{\theta_i} = v_{h_j} = 5$. We also assume that $S_{\theta_i} = 0.01^2$ for the coefficients of the lagged variables and $S_{h_j} = 0.1^2$ for the intercepts. Additionally, we fix $S_{h_j} = 0.1^2$. The priors for restricted models follow the same criteria according to the restrictions imposed on them.

4.3. Identification strategy

The identification of the VAR structural shocks is recursive and based on the Price Distribution Chain (PDC) model proposed by McCarthy (2007), which is standard approach in ERPT literature; see Maertens Odría et al. (2012) and Winkelried (2014), among others. The PDC model contributes to the analysis of the effect of financial, exchange rate, external and domestic demand shocks on each PDC stage. It also predicts that, given the nominal rigidities in the economy, the ERPT should decline along the distribution chain.

In this regard, we are considering the following ordering: $\mathbf{y}_t = [FED_t, XPI_t, GDP_t, DMP_t, NER_t, IPI_t, PPI_t, CPI_t]'$. Therefore, structural shocks associated with FED, XPI, GDP and DMP are associated with financial, external and domestic demand shocks, respectively. Likewise, the shocks associated with the NER are depreciation shocks of the PEN/US Dollar exchange rate. Finally, the shocks associated with IPI, PPI and CPI are domestic supply shocks along the distribution chain (import, producer, and consumer), respectively.

This ordering implies that financial, external and domestic demand, and exchange rate shocks have contemporaneous effects on the three domestic prices; and that the exchange rate triggers a contemporaneous effect on PDC; and that import and producer prices have an immediate effect on consumer prices, but not vice versa; see McCarthy (2007).

4.4. Evidence of time-varying parameters

Based on the estimation of the TVP-VAR-SV model, we employ the Kolmogorov-Smirnov and the t-test to establish whether the distribution and mean of the parameters at different points in time are the same, respectively. We apply both statistics to two cases. In the first, the sample is divided into 1995Q2-2008Q1 (pre-GFC period) and 2008Q4-2022Q4 (post-GFC period). In the second, the sample is divided into 1995Q2-2001Q4 (pre-IT period) and 2002Q1-2022Q4 (post-IT period).

Table 1 shows the results. We observe that a high percentage of parameters are time variants and that there is complete variability in the variance of the shocks. These results suggest that the estimation of a CVAR model could be inadequate as well as a model where everything is variable in time (TVP-VAR-SV).

4.5. Model selection

For each model we obtain a total of 100,000 posterior draws from 10 parallel chains after a burn-in of 1,000 in every chain, and keep every 10th draw for 10,000 final posterior draws which are used to estimate the log-ML_{CE} (based on importance sampling) and the DIC. The number of lags is 1, according to the BIC obtained from estimating the CVAR model.

Table 2 shows the ranking for the seven models. The log-ML_{CE} shows that the best-fitting model is the TVP-VAR-R3-SV model while the CVAR-SV model is quite close to the preferred model. At the same time, the DIC confirms the choice of the best model. In order to save space and gain concreteness, we chose the TVP-VAR-R3-SV and CVAR-SV models as the best. However, we include the TVP-VAR-SV and CVAR models as representative of a fully variable specification over time and another where constancy over time is imposed. Moreover, we highlight the importance of the SV component in the specification of the models, as the least-fitting ones (TVP-VAR and CVAR) do not include SV.

Fig. 2 shows the evolution of the SV over time for each equation for the selected models. The SV associated with the external variables (FED, XPI, NER, and IPI growth) shows an increasing trend from the beginning of the sample, reaches a peak during the GFC, declines in the following years to increase again during the last years of the sample (Russia-Ukraine war and pandemic COVID-19). Regarding the SV for the domestic variables, we observed a decreasing trend in the cases of PPI and CPI. GDP, after a period of relatively high volatility in the late 1990s, shows a clear moderation. Likewise, the domestic interest rate shows a great moderation



Fig. 1. Time Series.

Table 1
Tests for Time Variation in Coefficients and Volatility.

Kolmogorov-Smirnov test				
Pre and Post GFC			Pre and Post Inflation Targeting	
1995Q3-2008Q3 26/28	2008Q4-2022Q4 26/28	γ_t	1995Q3-2001Q4 26/28	2002Q1-2022Q4 26/28
1995Q3-2008Q3 58/72	2008Q4-2022Q4 61/72	β_t	1995Q3-2001Q4 56/72	2002Q1-2022Q4 61/72
1995Q3-2008Q3 8/8	2008Q4-2022Q4 8/8	h_t	1995Q3-2001Q4 8/8	2002Q1-2022Q4 8/8
t-test				
Pre and Post GFC			Pre and Post Inflation Targeting	
1995Q3-2008Q3 28/28	2008Q4-2022Q4 28/28	γ_t	1995Q3-2001Q4 28/28	2002Q1-2022Q4 27/28
1995Q3-2008Q3 55/72	2008Q4-2022Q4 58/72	β_t	1995Q3-2001Q4 53/72	2002Q1-2022Q4 58/72
1995Q3-2008Q3 8/8	2008Q4-2022Q4 8/8	h_t	1995Q3-2001Q4 8/8	2002Q1-2022Q4 8/8

γ_t represents the coefficients of contemporaneous relationships, β_t represents the coefficients associated to intercepts and lagged variables, and h_t represents the coefficients associated to volatility. In the fraction expression, the numerator indicates the number of parameters that vary and the denominator indicates the total number of parameters.

Table 2
Models Selection.

Model	log-ML _{CE}	Rank	DIC	Rank
TVP-VAR-SV	-1,654.28 (0.496)	4	2,723.49 (1.066)	4
TVP-VAR	-1,869.97 (0.447)	6	3,008.37 (1.020)	5
TVP-VAR-R1-SV	-1,645.53 (0.906)	3	2,597.66 (0.293)	2
TVP-VAR-R2-SV	-1,660.32 (0.916)	5	2,678.28 (1.148)	3
TVP-VAR-R3-SV	-1,639.96 (0.310)	1	2,587.11 (0.601)	1
CVAR-SV	-1,642.87 (0.176)	2	3,240.51 (18.236)	7
CVAR	-1,908.49 (0.042)	7	3,160.03 (0.009)	6

For each model, we obtain a total of 100,000 posterior draws from 10 parallel chains after a burn-in of 1,000 in every chain, and keep every 10th draw for 10,000 final posterior draws. log-ML_{CE} estimates are based on 10,000 evaluations of the integrated likelihood, where the importance sampling density is constructed using the 10,000 posterior draws. DIC estimates are computed using 10 parallel chains. In each chain, the integrated likelihood is evaluated for the 1,000 posterior draws kept from each estimation chain, getting a total of 10,000 evaluations. Standard deviations are presented in parentheses. Values in bold denote preferred models.

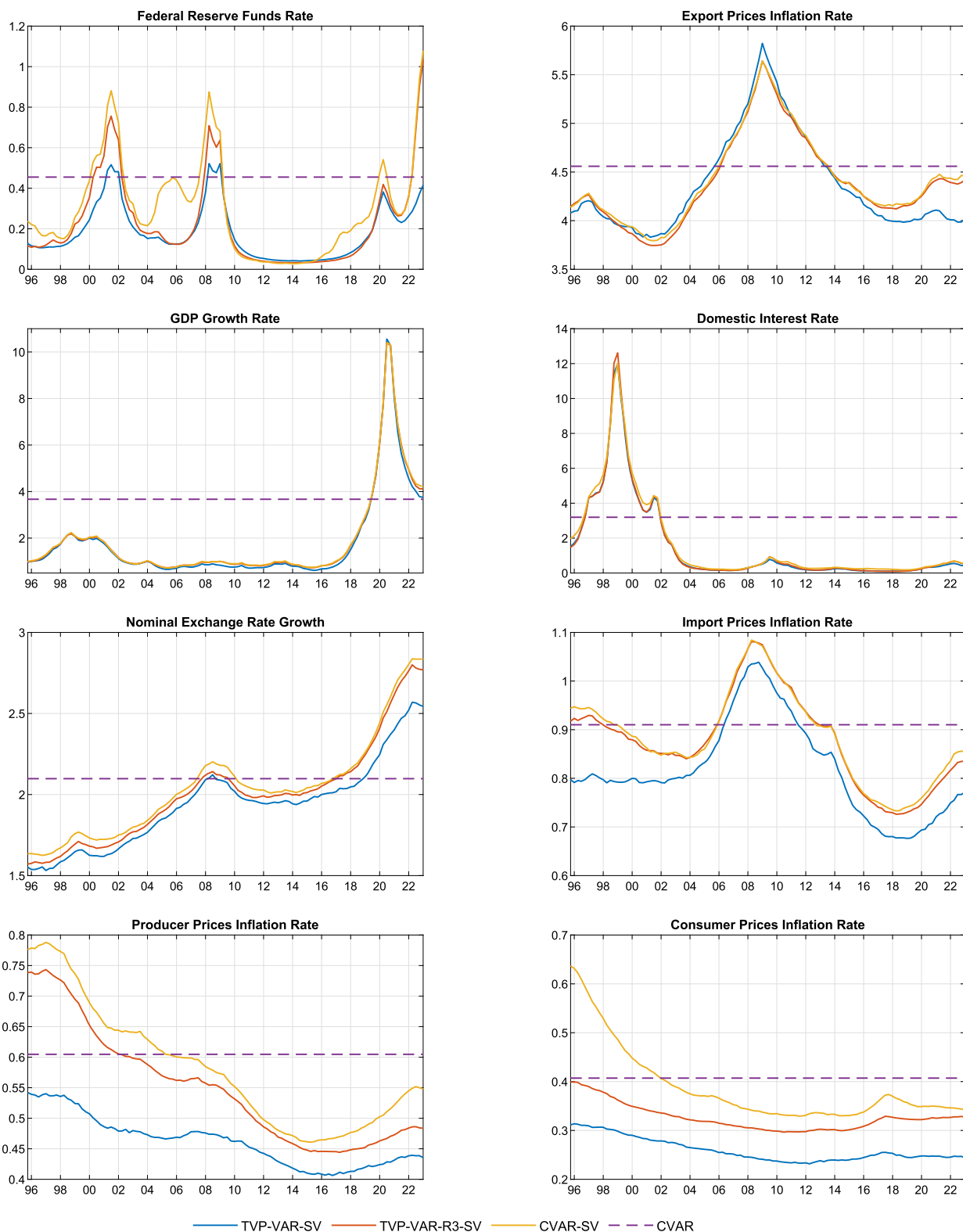


Fig. 2. Standard Deviation of the Innovations in each Equation for selected Models, Median Values. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Table 3
Median Long Run ERPT into Prices ($s = 20$).

Model	Import Prices	Producer Prices	Consumer Prices
TVP-VAR-SV	0.880 [0.758, 0.983]	0.560 [0.482, 0.701]	0.141 [0.118, 0.165]
TVP-VAR-R3-SV	0.825 [0.760, 1.130]	0.518 [0.430, 0.716]	0.114 [0.095, 0.167]
CVAR-SV	0.740 [0.697, 0.933]	0.458 [0.394, 0.581]	0.209 [0.183, 0.251]
CVAR	0.739 [0.739, 0.739]	0.455 [0.455, 0.455]	0.238 [0.238, 0.238]

Median ERPTs are calculated as the median of the evolution of ERPT rates over horizon $s = 20$ quarters and over time from 1995Q3 to 2022Q4. Confidence intervals are presented in brackets.

from 2002.¹⁴ These variables -with the exception of the interest rate- show a high volatility towards the final years of the sample, which is associated with the increase in inflation at the international level, the Covid-19 pandemic and political uncertainty related to the presidential elections in 2021.

4.6. Estimation of time-varying ERPT

We calculate the ERPT as the ratio of the cumulative impact over a time horizon (s) on domestic inflation (i = import, producer and consumer) of a 1% depreciation of the domestic currency divided by the cumulative impact over the same time horizon of the same shock into the growth rate of the nominal exchange rate. For each period the median of the simulations is considered. Thus, the ERPT for period t at time horizon s is defined as follows:

$$ERPT_i^s = \frac{\sum_{\tau=0}^s \frac{\partial PI_{i,t+\tau}}{\partial \epsilon_t}}{\sum_{\tau=0}^s \frac{\partial NER_{t+\tau}}{\partial \epsilon_t}}.$$

The ERPT is calculated separately for each price index, following the transmission mechanism described in McCarthy's (2007) PDC model. We consider a maximum time horizon of 20 quarters (s), and the median of the simulations is used for each period.

Table 3 displays the long-run median ERPTs for the best-fit TVP-VAR-R3-SV and CVAR-SV models, as well as for the TVP-VAR-SV and CVAR models, which is often used as a baseline in the literature. It should be noted that the median ERPTs differ somewhat across models due to their different specifications. Generally, we find that the TVP-VAR-SV and TVP-VAR-R3-SV models yield larger median ERPTs into import and producer prices than the CVAR-SV and CVAR models. Specifically, for the TVP-VAR-R3-SV model, the median ERPTs for import and producer prices are between 0.76-1.13 and 0.43-0.72, respectively, while, for the CVAR-SV model, those median ERPTs are between 0.69-0.93 and 0.39-0.58, respectively. In the case of the ERPT into consumer prices, the TVP-VAR-R3-SV model yields a median value of 0.10 for the ERPT into consumer prices, whereas the other models estimated larger values, closer to 0.2.

Fig. 3 displays the median of the three-dimensional plots of the evolution of rates of ERPT calculated for a 20-quarter horizon in the entire sample. The results are consistent with McCarthy's (2007) PDC model, which predicts a decline in the ERPT along the distribution chain due to factors such as price rigidities, market structure, and the composition of the basket of goods. Our best-fitting models show that the highest ERPT is observed in import prices, with a median value ranging from 0.76 to 1.13. Producer prices exhibit the second-highest ERPT, with a median value ranging from 0.43 to 0.72, followed by consumer prices, with a median value ranging from 0.1 to 0.17. These findings suggest that the transmission of exchange rate movements to prices is the strongest for import prices, followed by producer prices and consumer prices (see Fig. 4).

Moreover, Fig. 5 reveals an interesting finding: the estimated ERPTs into import and producer prices are higher in the short-term (4 quarters) than in the long-term (more than 20 quarters). Specifically, for import prices, the median short-term ERPT is approximately 1.2, while the long-term ERPT is close to 1.0. For producer prices, the short-term ERPT reaches 0.85, while the long-term ERPT is around 0.65. On the other hand, the ERPT into consumer prices shows a small difference between short and long-term, both being close to 0.1. These findings indicate that the transmission of exchange rate movements to prices varies depending on the time horizon and the type of price index considered.

Although we are unable to quantitatively assess the impact of price dollarization on ERPT as there is not time series of price dollarization for Peru, we recognize its potential significance and offer it as a plausible explanation for the observed patterns.¹⁵

¹⁴ This behavior is associated with macroeconomic stabilization policies in the mid-1990s, IT adoption, and sound monetary and fiscal policy implementation in recent decades; see Castillo et al. (2007). Additionally, Castillo et al. (2016) identify a significant attenuation in output and inflation volatility since the beginning of the 2000s (which they call the Great Moderation of Peru's economy). Primiceri (2005) and Cogley and Sargent (2005) find similar evidence for the US.

¹⁵ The link between ERPT and price dollarization is theoretically established by Drenik and Perez (2021).

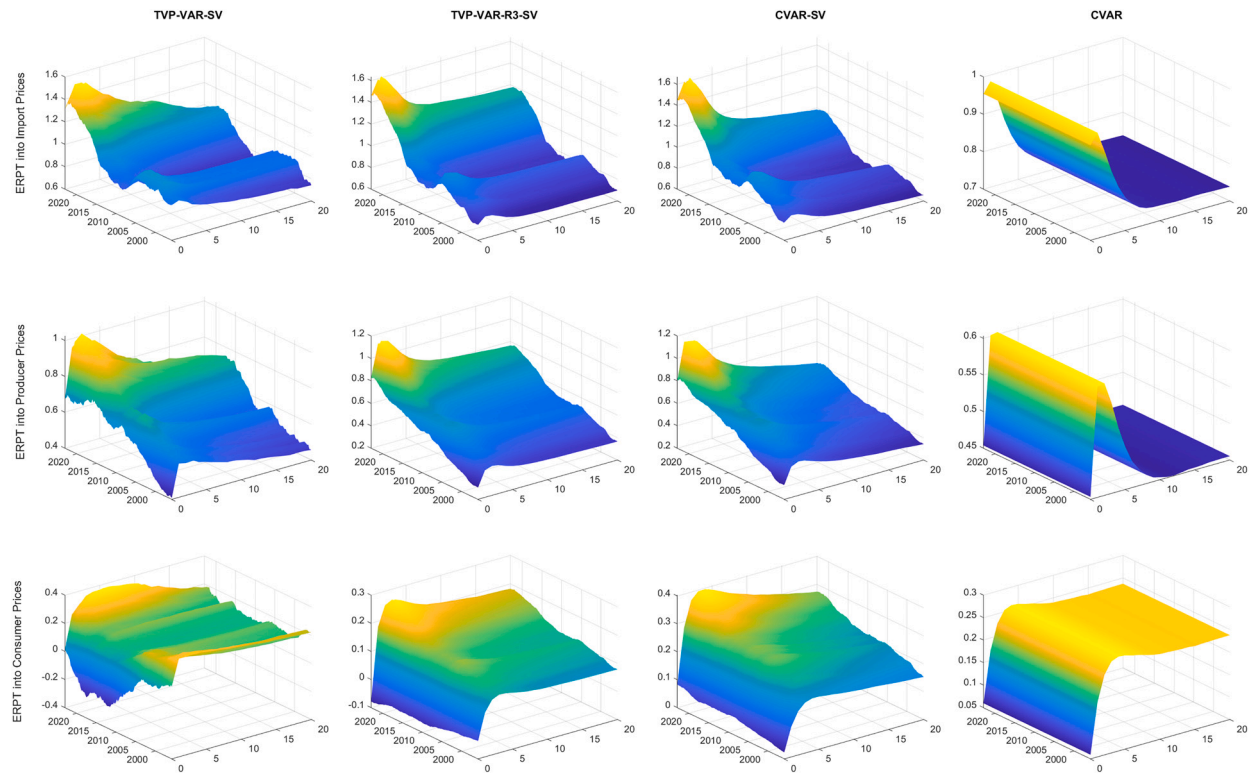


Fig. 3. Three-dimensional plots of the evolution of rates of ERPT into Import Prices (first row), Producer Prices (second row) and Consumer Prices (third row), over horizon $s = 20$ quarters and over time from 1995Q3 to 2022Q4, for selected Models. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Given the prevalent role of price dollarization in emerging economies, it is reasonable to explore its influence on the differential behavior of ERPT across price indexes. Consumer prices in Peru have very low-price dollarization, less than 5%, whereas price dollarization for producer prices may reach up to 40% and import prices close to 60%. Therefore, the differences in the ERPT might be attributed to the different degrees of price dollarization in import, producer, and consumer prices.

Furthermore, higher levels of price dollarization can offer an explanation for the observed discrepancy between short-term and long-term ERPT. In economies where prices for local consumers are denominated in foreign currency and prices in dollars remain inflexible, any change in the exchange rate has an immediate impact on local prices. However, as the demand for these goods and services is influenced by local factors, the relative price increase resulting from exchange rate fluctuations leads to a decrease in demand and, ultimately, a decline in prices denominated in US dollars. This adjustment process occurs gradually, which elucidates the decline of ERPT from its short-term value to its long-term value over a span of 20 quarters, as depicted in Fig. 5. This finding is significant as it demonstrates how price dollarization can influence the dynamics of ERPT.

From a policymaker's perspective, these findings imply that, in dollarized economies, inflation forecasts should account for the fact that short-term ERPT tends to be higher than long-term ERPT, taking into consideration the time lag for full adjustment. Additionally, it is important to consider the varying degrees of price dollarization across different price aggregates. By recognizing these nuances, policymakers can develop more accurate inflation forecasts and make informed decisions to manage the impacts of ERPT in dollarized economies.

4.7. ERPT evolution

Fig. 6 shows the evolution of the long-term ERPT (20 quarters) for the three models with the best fit and the CVAR model as benchmark for each aggregate price level separately (Panel A) and for the evolution of the three ERPTs for each model (Panel B).

After reaching a minimum value in 2008/2009, the ERPT into import and producer prices has exhibited persistent increases, particularly after 2014. In contrast, the ERPT into consumer prices demonstrates less variability, remaining below 20 percent.

Interestingly, despite the increasing ERPT into import and producer prices, there has been a continuous decrease in the volatility of inflation for these two price aggregates until 2020. However, in the last two years of the sample, the volatility of inflation has shown an upward trend, which can be attributed to the significant rise in international prices of energy and soft commodities (see Fig. 2). The reduction in volatility of import and producer inflation, despite the larger ERPT, suggests that other factors, such as domestic aggregate demand and idiosyncratic supply factors, have become more stable over time. This has contributed to maintaining

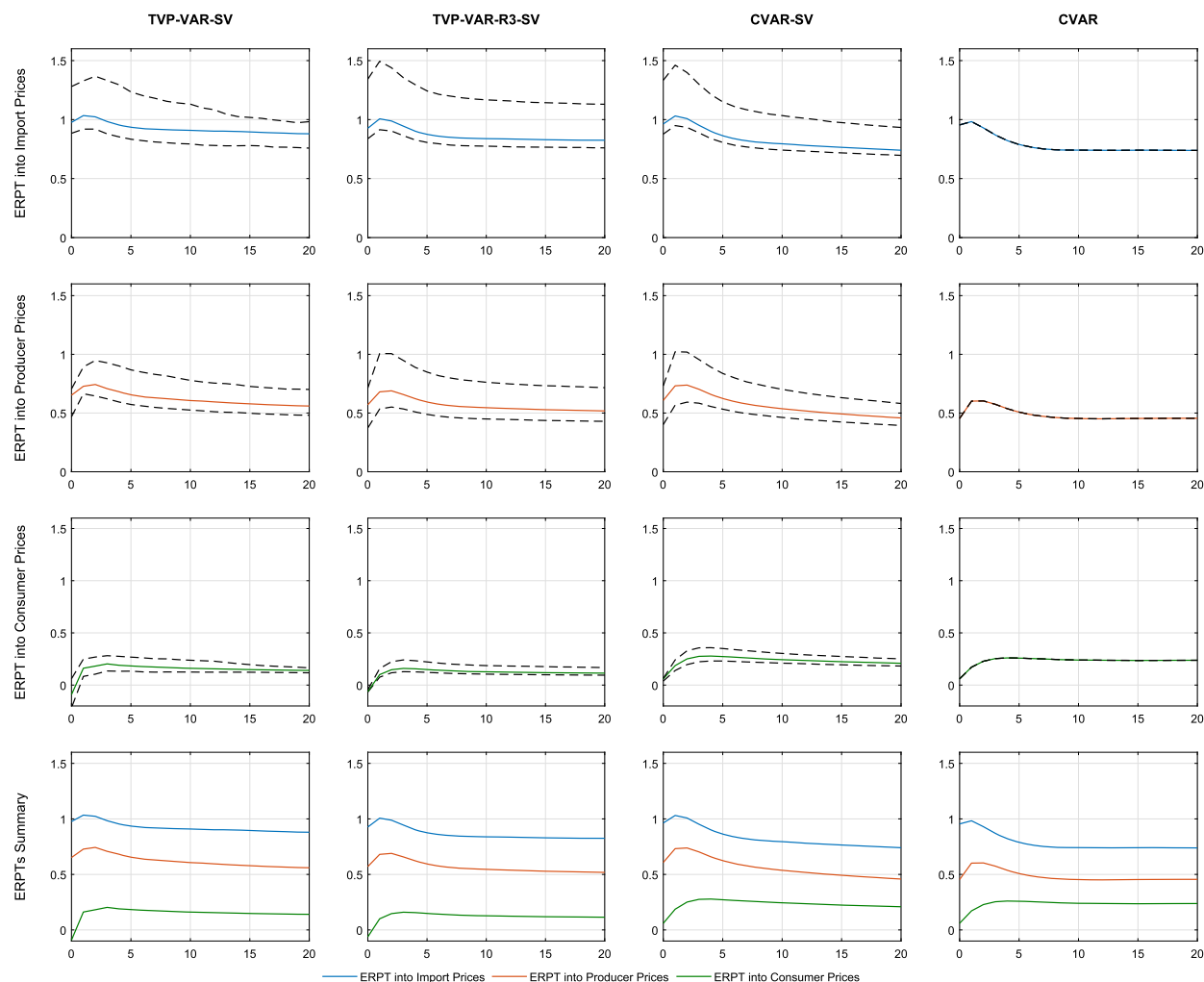


Fig. 4. Median ERPT into Import (first row), Producer (second row) and Consumer (third row) Prices. The black lines represent the 68% error bands. The fourth row presents the ERPTs summary for selected Models. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

overall stable inflation rates for these price indexes. Additionally, improved monetary policy, which has effectively reduced inflation uncertainty, may have played a role in keeping import and producer price inflation lower during the period spanning 2014 to 2020.

Also, Fig. 7 reveals that the median short-term ERPT into import and producer prices experienced a notable surge from the period 2011–2019 to the period 2020–2022, rising from 1.2 to 1.6 and from 0.85 to 1.15, respectively. However, the long-term ERPT exhibited a more modest increase, climbing from 1.0 to 1.2 for import prices and from 0.65 to 0.75 for producer prices. These results might suggest that an increasing number of firms have chosen to price their products in dollars since 2020, likely motivated by greater uncertainty surrounding exchange rates and macroeconomic policies in general. This trend, combined with the sharp upturn in oil and food prices during these years,¹⁶ may offer an explanation for the rapid surge in import and producer prices witnessed in Peru throughout 2021 and 2022.

Also, the evolution of the short-run and long-run ERPT since 2011 could be partially attributed to a greater number of firms consciously choosing to price their goods and services in dollars. Drenik and Perez (2021) offer both theoretical explanations and empirical evidence supporting this claim. According to their research, higher inflation rates, lower variance of exchange rate shocks, higher dollarization of household assets, a larger share of tradable inputs, and higher dollarization of firms' debt contribute to increased levels of price dollarization.

In the case of Peru, we have observed an upward trend in average inflation rates, rising from an average of 2.7% during the 2002–2013 period to an average of 3.7% during the 2014–2022 period. Additionally, inflation volatility has also increased from 1.7 to 2.4 between similar periods. Furthermore, although the volatility of exchange rate shocks has increased since 2014 (as depicted in Fig. 2), the level of dollarization in bank deposits has shown an upward trend. From 44% in December 2012, it reached 54%

¹⁶ See Cornejo et al. (2022) for a decomposition of the contribution of external shocks to domestic inflation in Peru.

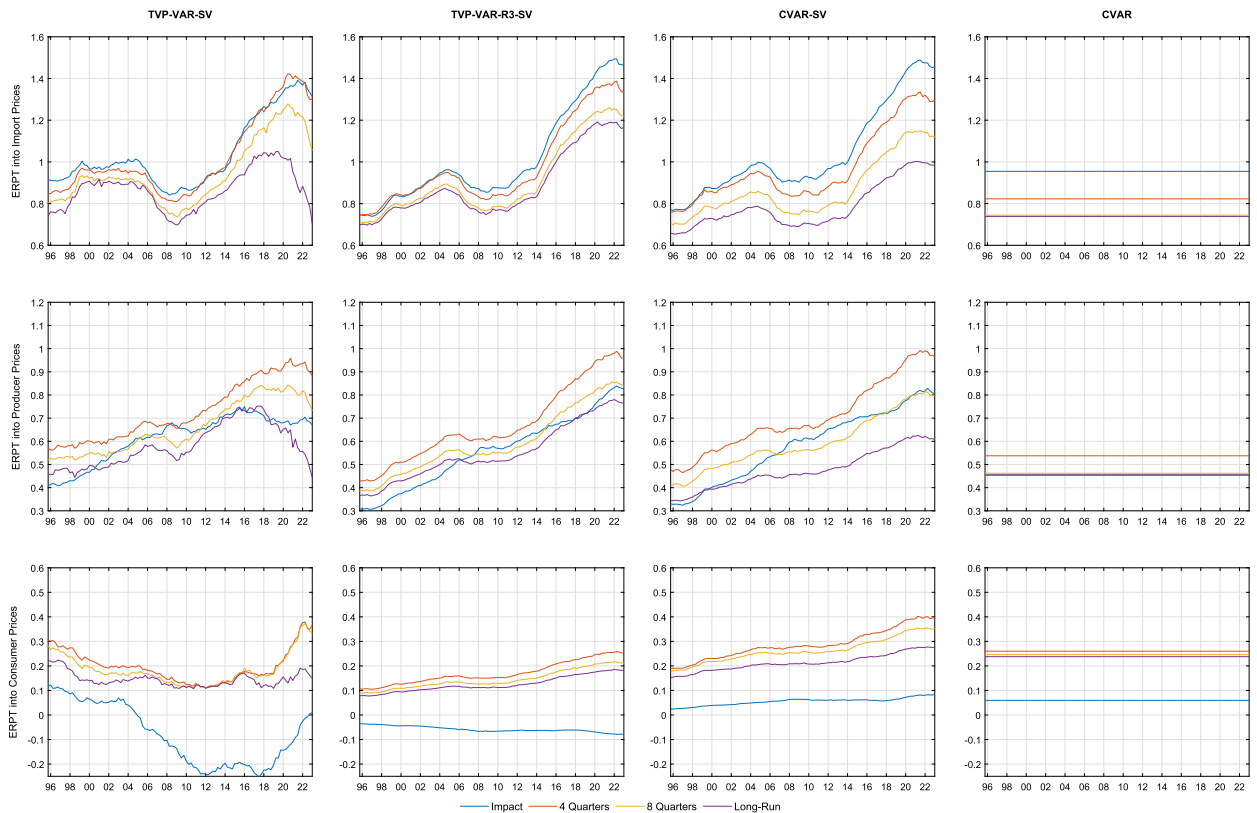


Fig. 5. Time-Varying Impact ERPT, 4, 8 quarters and Long Run ERPT into Import (first row), Producer (second row) and Consumer (third row) Prices for selected Models. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

in February 2016, coinciding with a period of rapid strengthening of the dollar. Since 2016, dollarization levels have declined but remained above 42%. The incentives for firms to adopt foreign currency pricing have become even more pronounced since 2021 due to heightened political uncertainty and elevated levels of inflation that have impacted the economy during these years.

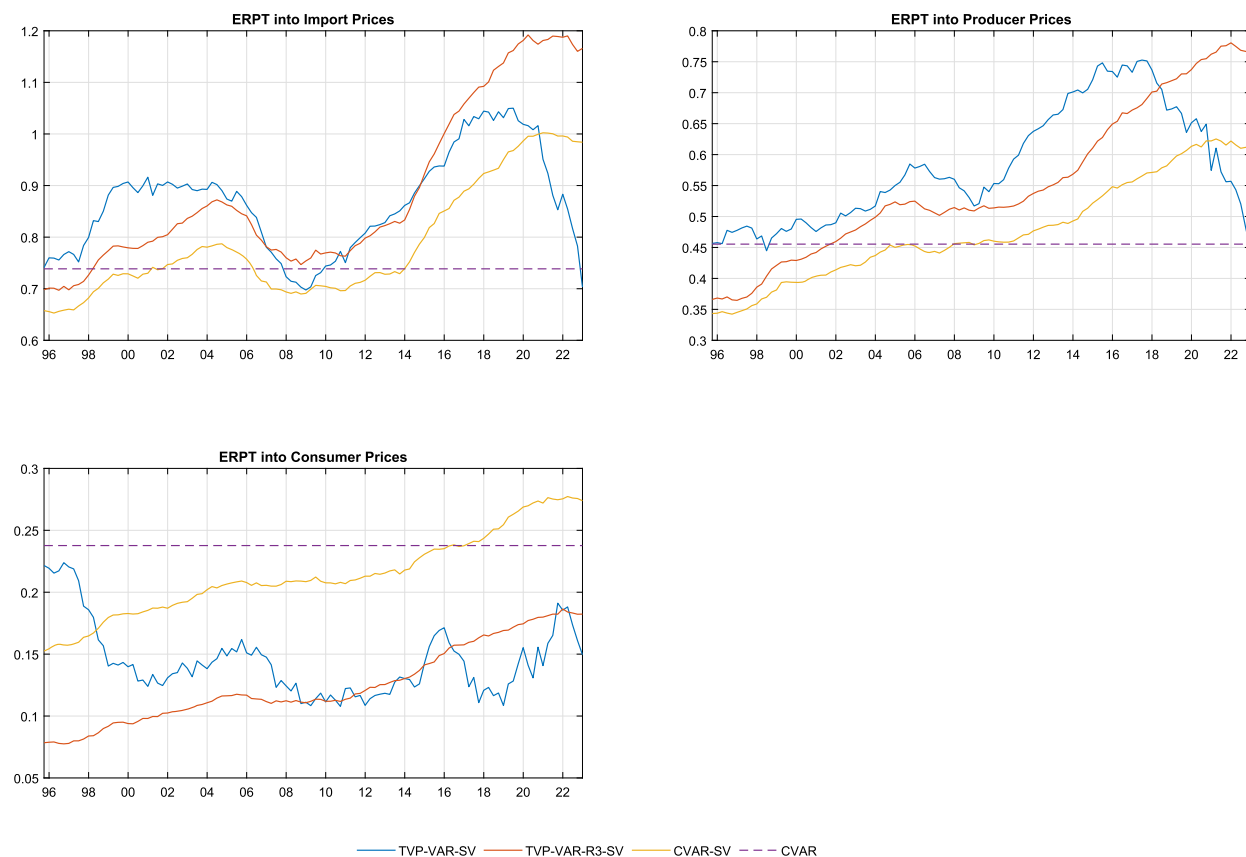
In summary, our findings reveal the following key aspects of ERPT: (i) the ERPT exhibits temporal variation, indicating that its magnitude changes over time, (ii) the ERPT diminishes along the distribution chain, suggesting that the impact of exchange rate fluctuations becomes less pronounced as prices move from import to producer to consumer levels, (iii) the long-term ERPT is lower than the short-term ERPT, which aligns with the presence of higher levels of price dollarization in import and producer prices, (iv) since 2009, there has been an increase in ERPT into import and producer prices, while consumer prices have experienced relatively low ERPT, possibly reflecting higher levels of price dollarization for import and production prices, and (v) notably, the short-term ERPT into import and producer prices has undergone the most significant increase since 2020, potentially reflecting the influence of heightened political and macroeconomic uncertainty during this period.

4.7.1. ERPT into import prices

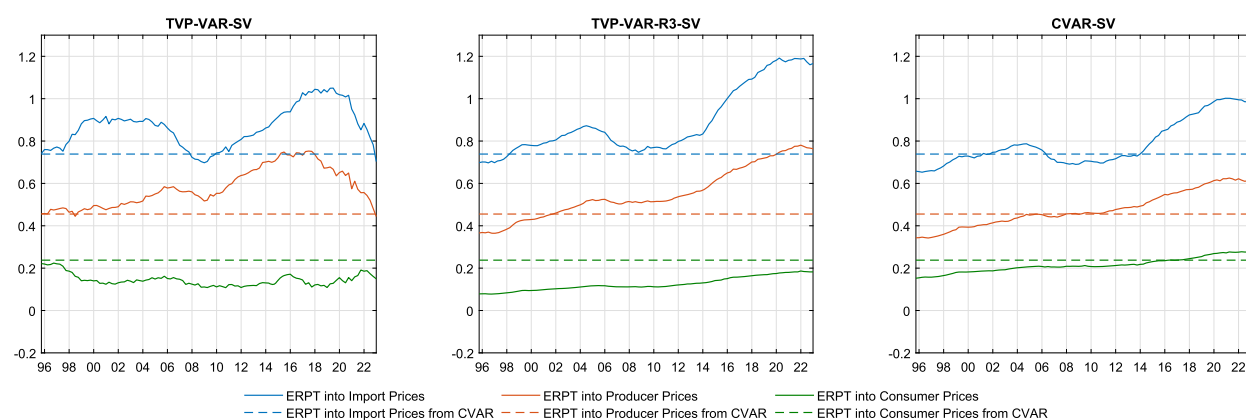
Fig. 6 (Panel A) shows the evolution and magnitude of ERPT into import prices for the best fitting models and for the CVAR. The estimation of the ERPT at the beginning of the sample lies between 0.7-0.75; then increases towards the range of 0.8-0.90 by 2005. Between 2005-2009, the ERPT into import prices shows a reduction reaching its minimum values at the end of 2009. Our results show that, since 2009, the ERPT increases persistently, reaching by the end of the sample values close to 1.2. The ERPT into import estimated using the TVP-VAR-R3-SV are larger than those of the TVP-VAR-SV after 2014, whereas the opposite is observed before 2009.

In line with our results, Gondo and Pérez-Forero (2019) provide evidence of a resurgence of the ERPT into import prices after the GFC, when EMEs experienced considerable capital inflows from advanced economies due to low international interest rates and EMEs solid macroeconomic fundamentals. Higher volatility in capital flows increases the volatility of the exchange rate, which can generate additional incentives for importing firms to set prices for domestic customers in dollars. For these firms, invoicing in dollars is a form of exchange rate hedging, as a large fraction of their cost are in dollars, which become more important as the exchange rate volatility increases. This explanation is further supported by the evolution of the short-term ERPT into import prices, which has increased more rapidly than the long-term ERPT.

This finding is consistent with recent literature on ERPT, which highlights an increase in ERPT in advanced economies during and after the GFC due to expansionary monetary policies such as Quantitative Easing and the taper tantrum. Several studies employing



Panel A. Time-Varying Long Run ERPT into Import, Producer and Consumer Prices for Selected Models Grouped by Price Type



Panel B. Time-Varying Long Run ERPT into Import, Producer and Consumer Prices Grouped by Selected Models

Fig. 6. Evolution of Long Run ERPT into Prices. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

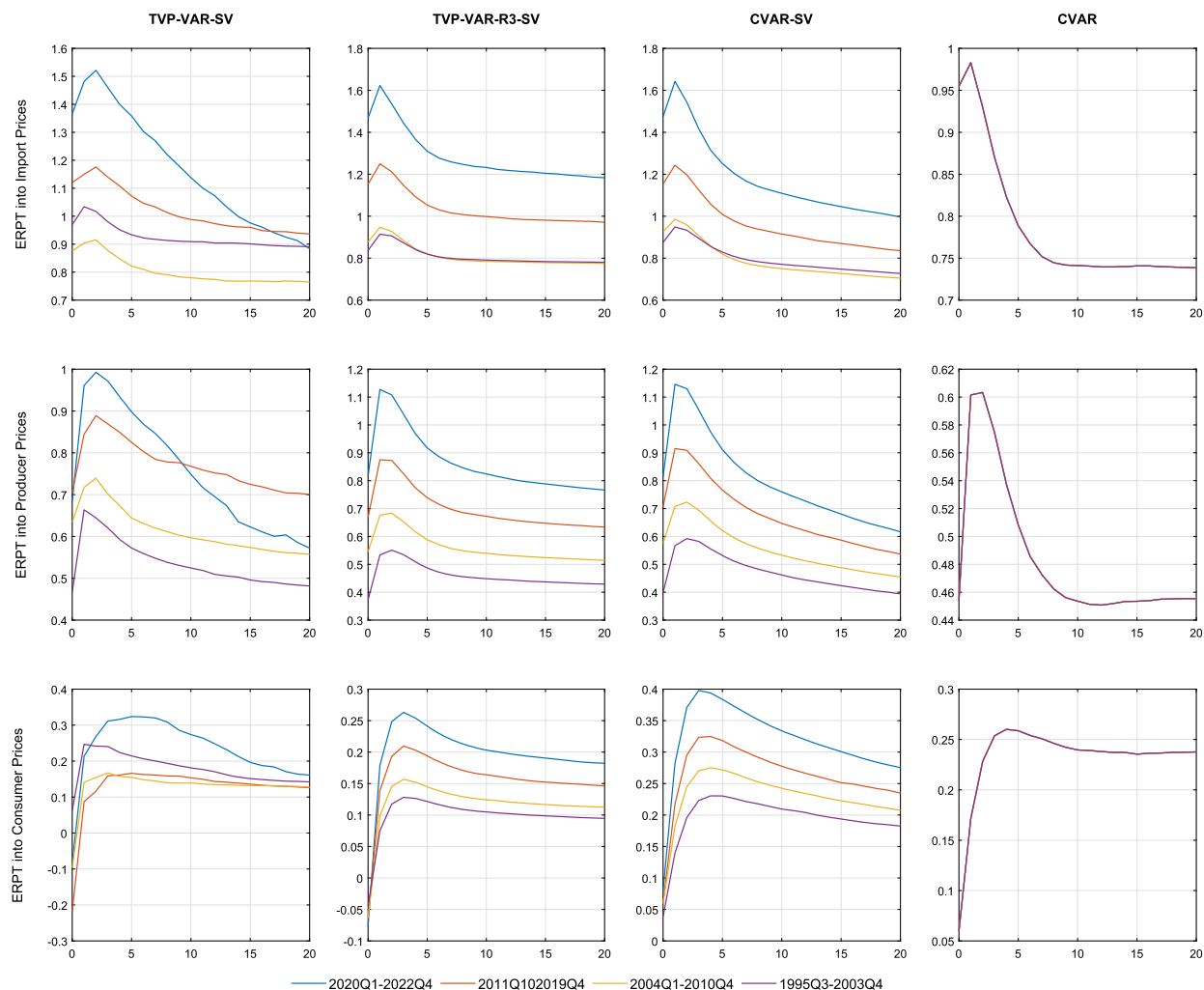


Fig. 7. Evolution of Long Run ERPT into Prices. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

TVP-VAR-SV models, such as Ozkan and Erden (2015) and Alexius and Holmberg (2017), have empirically supported this trend. They demonstrate that ERPT, which had declined in advanced economies during the 1990s, has since increased, particularly during periods of significant depreciation of domestic currencies against the dollar. A separate study conducted by Forbes et al. (2018) found similar evidence for the UK, utilizing a different methodology.

Alexius and Holmberg (2017) point out that FX markets in advanced economies have remained volatile since the GFC, accompanied by an increase in inflation. However, evidence regarding ERPT in EMEs is still limited. Jooste and Jhaveri (2014) examine South Africa, while Dahem et al. (2017) focus on Tunisia, both presenting evidence of a substantial surge in ERPT during and after the GFC.

Interestingly, recent studies such as Shioji (2014), Sasaki and Yoshida (2018), Sasaki (2019) and Sasaki et al. (2022) showed that the ERPT in Japan started to increase around the time of the GFC, in a context of low inflation, similar to our findings for Peru. Recent changes in ERPT behavior in Japan are attributed mainly to increased co-movement between the currency and the price index as well as the changes in firm pricing behavior (Hara et al., 2015) and to higher volatility of the Yen and that of the price of oil.

4.7.2. ERPT into producer prices

In Fig. 6 (Panel A), it is evident that all models demonstrate an upward trend in ERPT following the GFC. The estimated best-fitting model indicates an initial increase in ERPT into producer prices, rising from 0.35 to 0.50. Subsequently, between late 2005 and mid-2008, it remained relatively stable around 0.5. However, similar to the ERPT into import prices, a notable resurgence in ERPT into producer prices occurred, reaching a peak value of 0.78 in 2020. Since then, ERPT has experienced a slight decline, hovering slightly above 0.75.

The CVAR-SV model showcases a similar pattern for the ERPT into producer prices but with maximum values of 0.5, which is lower than the estimated value reported by the best-fitting model (TVP-VAR-R3-SV). On the other hand, the estimation of the ERPT

into producer prices obtained from the TVP-VAR-SV model differs somewhat from the previous two models. It reveals an increase from 2009 onwards, peaking at 0.65 in 2020. However, in the last two years, ERPT declines to 0.45. Since this model does not align as well with the two selection criteria used, we consider the results of this model to be complementary rather than definitive.

The increasing trend in the ERPT into producer prices can be attributed to the evolving pattern of price invoicing in the dollar denomination within this sector. This observation is supported by Contreras et al. (2017), who present evidence regarding the persistent dollarization of producer prices. Their findings align with our own, suggesting that higher levels of price dollarization can contribute to an amplified ERPT. Using data from the 2015–2016 BCRP Macroeconomic Expectations Survey, Contreras et al. (2017) demonstrates the persistence in levels above 40% of dollarization of transactions and production costs in Peru. According to their study, the dollarization of input costs for non-financial firms is high, at around 54%, and the financial costs of 44% of these firms are denominated in dollars. They also note that high levels of dollarization of input sales and purchases make it challenging to replace foreign currency with domestic currency, given the rigidity in price and transaction dollarization within Peru's economy. Contreras et al. (2017) further find that dollarization of administrative expenses is particularly high in the mining and fishing industries. Rossini et al. (2016) provide similar evidence to support these findings.

Regarding the dollarization of sales by Peruvian non-financial firms, Contreras et al. (2017) report that most durable goods, such as real estate and cars, are typically negotiated in dollars, with 55% of these firms fixing their sales contracts in dollars. In addition, around 92% of sales by export-oriented industries, such as mining and industrial fishing, are denominated in dollars. Among other industries, dollar sales constitute 72.3% for transportation/communications, 65% for export agriculture, 56.7% for commerce, 52% for services, and 27.9% for construction; however, dollar sales are less prevalent in energy industries. The authors also note that approximately 43% of non-financial firms show a negative mismatch, indicating that their dollar liabilities exceed their dollar assets, which makes them vulnerable to exchange rate fluctuations.

4.7.3. ERPT into consumer prices

Consistent with the predictions of the PDC model, the ERPT into consumer prices exhibits the lowest values among the three calculated ERPT measures. Fig. 6 (Panel A) illustrates that both the TVP-VAR-R3-SV and CVAR-SV models demonstrate a similar time evolution for the ERPT into consumer prices. In the case of the best-fitting model, the ERPT starts at 0.1 at the beginning of the sample period and gradually increases, reaching a maximum of 0.2 by the end of the sample. The CVAR-SV model exhibits a similar pattern, albeit with higher values.

In contrast, the TVP-VAR-SV model presents a different trend. The ERPT into consumer prices initiates at approximately 0.2 at the beginning of the sample period, but experiences a decline to below 0.1 in 2009. From 2009 onwards, there is an upward trend until 2019, followed by a slight regression to 0.2 at the end of the sample period.

It is interesting to note that the short-term ERPT and the long-term ERPT show also the lowest difference in comparison with the same difference for import and producer prices (Fig. 7). This is consistent with the fact that price dollarization is the lowest among the three price indexes analyzed, consumer, import prices and producer prices. There are several explanations for why price dollarization is low for consumer prices. On one hand, Castillo and Montoro (2016) show that only luxury goods have a higher probability of having prices in dollars, whereas necessity goods are invoiced in domestic currency. As the participation of luxury goods is small in the measurement of consumer prices, price dollarization tends to be low as well. Similarly, there is some empirical evidence that shows that large ticket items have prices in dollars, whereas small ticket items are in domestic currency, which can be associated to the cost of having small change in foreign currency.¹⁷

From a theoretical perspective, Maertens Odría et al. (2012) argue that, after IT adoption, the BCRP prioritized inflation volatility over exchange rate fluctuations. This resulted in higher exchange rate volatility, in turn lowering the share of domestic firms fixing their prices in dollars and reducing the ERPT into consumer prices. From another point of view, Montoro (2006) argues that, after implementation of Law 28300 (which compelled firms to establish their prices in Soles), domestic firms have a disincentive to establish their prices in foreign currency, as they incur in menu costs when adjusting their prices in response to exchange rate variations. In this line, at a more disaggregated level, Castellares (2017) and Castellares and Toma (2020) show evidence of a decreasing ERPT into consumer prices after implementation of Law 28300. In particular, Castellares and Toma (2020) show that the ERPT into non-durable goods drops to 0.19 and practically disappears after implementation of the Law.

It is worth to mention, however, that the time evolution of ERPT into consumer prices after the adoption of the IT regime in 2002 differs across the best fitting models. For the estimations obtained from the CVAR-SV and TVP-VAR-R3-SV models, the ERPT did not fall after the adoption of the IT regime in Peru.

Finally, Panel B of Fig. 6 complements the analysis of the evolution and magnitude of the ERPT, showing that for any selected model, the ERPT of import prices exceeds that of producer prices, and the latter, in turn, exceed the ERPT of consumer prices throughout the sample horizon.

5. Robustness analysis

In order to validate the results obtained from the baseline model, we perform four robustness exercises: (i) estimation of baseline model with Lenza and Primiceri (2022) Covid-19 pandemic adjustment; (ii) changing the Federal Reserve Funds Rate for the shadow

¹⁷ In many countries, like Peru and Uruguay, prices of car and real state, big ticket items are invoiced in US dollars, whereas prices of small ticket goods, such as food and perishable goods are invoiced in domestic currency.

Table 4
Models Selection.

Model	Robustness 1				Robustness 2				Robustness 3			
	log –ML _{CE}	Rank	DIC	Rank	log –ML _{CE}	Rank	DIC	Rank	log –ML _{CE}	Rank	DIC	Rank
TVP-VAR-SV	-1,623.35 (0.481)	5	2,654.43 (0.551)	4	-1,686.29 (0.640)	3	2,788.47 (0.535)	4	-1,735.88 (0.252)	4	2,879.14 (1.063)	4
TVP-VAR	-1,758.22 (0.302)	6	2,887.28 (1.584)	6	-1,890.84 (0.194)	6	3,064.76 (1.491)	5	-1,974.34 (0.702)	6	3,283.61 (0.258)	5
TVP-VAR-R1-SV	-1,602.62 (0.971)	3	2,544.22 (1.336)	2	-1,697.35 (0.719)	5	2,735.40 (0.183)	2	-1,717.92 (0.705)	3	2,737.10 (0.421)	2
TVP-VAR-R2-SV	-1,615.05 (0.901)	4	2,625.92 (1.700)	3	-1,678.08 (0.606)	1	2,742.40 (0.393)	3	-1,738.67 (1.231)	5	2,824.01 (1.347)	3
TVP-VAR-R3-SV	-1,597.09 (0.416)	1	2,543.37 (0.849)	1	-1,685.60 (0.307)	2	2,692.94 (0.352)	1	-1,710.14 (0.272)	1	2,717.04 (0.948)	1
CVAR-SV	-1,602.12 (0.161)	2	2,756.80 (14.436)	5	-1,694.96 (0.116)	4	4,165.41 (27.589)	7	-1,714.40 (0.156)	2	3,360.39 (7.893)	7
CVAR	-1,792.93 (0.062)	7	2,949.04 (0.001)	7	-1,935.99 (0.170)	7	3,216.97 (0.010)	6	-1,991.82 (0.089)	7	3,305.30 (0.018)	6

For each model, we obtain a total of 100,000 posterior draws from 10 parallel chains after a burn-in of 1,000 in every chain, and keep every 10th draw for 10,000 final posterior draws. log-ML_{CE} estimates are based on 10,000 evaluations of the integrated likelihood, where the importance sampling density is constructed using the 10,000 posterior draws. DIC estimates are computed using 10 parallel chains. In each chain, the integrated likelihood is evaluated for the 1,000 posterior draws kept from each estimation chain, getting a total of 10,000 evaluations. Standard deviations are presented in parentheses. Values in bold denote preferred models.

federal funds rate proposed by Wu and Xia (2016); (iii) changing the export prices inflation rate for the Global Price of Metal Index growth rate; and (iv) identification via sign restrictions scheme. The robustness exercises focus on estimating long-run ERPTs into the three types of domestic prices. All the figures related to the exercises are found in an Appendix available upon request.¹⁸

5.1. Baseline model with Lenza and Primiceri (2022) Covid-19 adjustment

Given that the data used presents an atypical behavior in the last period of the sample due to the Covid-19 pandemic, in this robustness exercise we adjusted for said anomaly according to the proposal of Lenza and Primiceri (2022). In particular, the adjustment is made for the quarters more deeply affected by the occurrence of the Covid-19 pandemic in Peru, from the second quarter to the fourth quarter of 2020. In general, it can be seen that the ERPT into prices do not change much with respect to what was found in the baseline model.

Table 4 (see column of Robustness 1) shows the two Bayesian selection criteria. The DIC and the log-ML indicate that the best models are the TVP-VAR-R3-SV and CVAR-SV models, which coincide with the choice in the baseline model.

Table 5 (column Robustness 1) shows the long-run ERPT values for selected models. If we focus on the long run ERPTs of the best fitting model, it shows that the adjustment by Lenza and Primiceri (2022) to variables from the baseline model tend to slightly increase the levels of ERPTs in all cases. However, it does not alter the evolution over time of the ERPT nor the ranking with respect to the baseline model.

In particular, we find that: (i) the ERPT into import prices, at the beginning of the sample, has a value of 0.70, exhibiting a significant increase until reaching a maximum value of approx. 0.87 in 2005. Afterwards, it exhibits a reduction until mid-2009, reaching a value of 0.78. During the 2009-2014 period, a relatively stable behavior is observed. From 2014 until the end of the sample, the ERPT shows a significant increase, reaching its maximum value of 1.21 at the beginning of 2022; (ii) the ERPT into producer prices at the beginning of the sample has a value of 0.39, and exhibits a growing trend throughout the analysis period. During the 2004-2011 period, the ERPT shows a relatively stable behavior with a value that fluctuates around 0.50. In the period post-GFC, a sustained increase is observed, reaching a value close to 0.80 at the end of the sample; (iii) the ERPT into consumer prices at the beginning of the sample has a value of 0.10, and exhibits a slight upward trend throughout the analysis period, reaching a value close to 0.20 at the end of the sample.

¹⁸ A preliminary version associated with Calero et al. (2022) includes several robustness experiments but using a model with 6 variables (there is no foreign or domestic monetary policy) and with a sample up to 2019Q4. The experiments are (i) change the first variable (XPI) in the ordering of the baseline VAR model by GPIAC, S&P GSCI, Copper Prices, Terms of Trade, US Consumer Inflation and US GDP Deflator Inflation, respectively; (ii) change the variable associated with domestic economic activity; (iii) establish the values of the priors based on a training sample; and (iv) the estimate of the ERPT for another Latin American economy (Colombia). This Appendix is also available upon request.

Table 5
Median Long Run ERPT into Prices ($s = 20$).

Model	Robustness 1			Robustness 2		
	Import Prices	Producer Prices	Consumer Prices	Import Prices	Producer Prices	Consumer Prices
TVP-VAR-SV	0.836 [0.723, 0.905]	0.533 [0.491, 0.659]	0.147 [0.118, 0.241]	0.942 [0.807, 1.162]	0.729 [0.615, 0.838]	0.208 [0.125, 0.257]
TVP-VAR-R2-SV				0.775 [0.668, 1.050]	0.582 [0.491, 0.668]	0.163 [0.143, 0.224]
TVP-VAR-R3-SV	0.838 [0.767, 1.126]	0.529 [0.434, 0.717]	0.142 [0.115, 0.190]	0.838 [0.731, 1.046]	0.547 [0.490, 0.719]	0.171 [0.152, 0.218]
CVAR-SV	0.749 [0.702, 0.937]	0.486 [0.416, 0.611]	0.244 [0.203, 0.284]			
CVAR	0.707 [0.707, 0.707]	0.421 [0.421, 0.421]	0.241 [0.241, 0.241]	0.696 [0.696, 0.696]	0.456 [0.456, 0.456]	0.283 [0.283, 0.283]

Model	Robustness 3			Robustness 4 (Sign Restrictions)		
	Import Prices	Producer Prices	Consumer Prices	Import Prices	Producer Prices	Consumer Prices
TVP-VAR-SV	0.759 [0.620, 1.094]	0.494 [0.411, 0.673]	0.099 [0.030, 0.224]	0.503 [0.442, 0.554]	0.397 [0.348, 0.467]	0.230 [0.197, 0.270]
TVP-VAR-R2-SV				0.431 [0.405, 0.463]	0.308 [0.285, 0.344]	0.179 [0.157, 0.223]
TVP-VAR-R3-SV	0.956 [0.861, 1.572]	0.672 [0.531, 1.004]	0.122 [0.104, 0.191]	0.451 [0.430, 0.502]	0.267 [0.250, 0.297]	0.129 [0.123, 0.137]
CVAR-SV	0.912 [0.836, 1.311]	0.615 [0.505, 0.771]	0.172 [0.152, 0.229]	0.444 [0.435, 0.476]	0.217 [0.196, 0.238]	0.167 [0.160, 0.179]
CVAR	0.691 [0.691, 0.691]	0.446 [0.446, 0.446]	0.203 [0.203, 0.203]	0.514 [0.514, 0.514]	0.352 [0.352, 0.352]	0.273 [0.273, 0.273]

Median ERPTs are calculated as the median of the evolution of ERPT rates over horizon $s = 20$ quarters and over time from 1995Q3 to 2022Q4. Confidence intervals are presented in brackets.

5.2. Baseline model with change of the external monetary policy variable

In this robustness exercise, the variable associated with the external monetary policy (Federal Reserve funds rate) is changed in the baseline model for the shadow federal funds rate as proposed by Wu and Xia (2016). According to the Bayesian selection criteria shown in Table 4 (see column Robustness 2), the two models with the best fit are the TVP-VAR-R2-SV and the TVP-VAR-R3-SV models.

In the case of the TVP-VAR-R3-SV model, we observe a slight reduction in the levels of long-term ERPTs. However, it does not alter the evolution over time of the ERPT nor the ranking with respect to the baseline model in a large part of the sample.

According to the Table 5 (column Robustness 2), we find that: (i) the ERPT into import prices, at the beginning of the sample, has a value of 0.74, exhibiting a slight increase until reaching a value of around 0.87 in 2005. Then, it exhibits a reduction until mid-2009, reaching a value of 0.70. During the 2009-2020 period, this ERPT shows a significant increase, reaching its maximum value around 1.10 at the 2018. Meanwhile, for the 2018-2022 period, a slight reduction is experienced, reaching a value close to 0.99 at the end of the sample; (ii) the ERPT into producer prices at the beginning of the sample has a value of 0.40, and exhibits a growing trend throughout the analysis period, reaching a value of around 0.72 in 2022. After 2016, this ERPT tends to remain stable until the end of the sample with a value around 1.00; (iii) the ERPT into consumer prices at the beginning of the sample has a value of 0.12, and exhibits a slight upward trend throughout the analysis period, reaching a value close to 0.22 at the end of the sample.

5.3. Baseline model with change of the external demand variable

There is abundant empirical evidence on the role of external shocks in the performance of emerging economies such as Peru, see for example, Rodríguez et al. (2018, 2023, 2024) and Chávez and Rodríguez (2023). Therefore, in line with Miller (2003), this robustness exercise consists of changing the variable associated with external demand in the baseline model (export prices inflation rate) for the Global Price of Metal Index growth rate. In this case (Table 4, column of Robustness 3), the Bayesian selection criteria indicate that the two models with the best fit are the TVP-VAR-R3-SV and the CVAR-SV models.

According to Table 5 (column Robustness 3), results for best models show significant increase in the levels of ERPTs into import and producer prices, but not the levels of the ERPT into consumer prices. However, it does not alter the evolution over time of the

ERPT nor the ranking with respect to the baseline model in a large part of the sample. In particular, it is found that: (i) the ERPT into import prices at the beginning of the sample has a value of 0.80, exhibiting a significant increase until reaching a maximum value of around 1.00 in 2005. Later, it exhibits a reduction until mid-2009, reaching a value of 0.82. During the 2009–2018 period, this ERPT shows a significant increase, reaching its maximum value around 1.70 in 2018. Meanwhile, for the 2018–2022 period there is a slight reduction, reaching a value close to 1.40 at the end of the sample; (ii) the ERPT into producer prices at the beginning of the sample has a value of 0.40, and exhibits a growing trend throughout the analysis period. From the beginning of the sample until 2016, the ERPT shows a sustained increase reaching a value around 1.10 in 2016. After 2016, this ERPT tends to remain stable until the end of the sample with a value around 1.00; (iii) the ERPT into consumer prices presents behavior and values similar to what was found in the baseline model.

In summary, this exercise provides similar evolution patterns for all ERPTs, although the magnitudes are higher than in the baseline model. These levels seem to be connected to the inclusion of a more volatile variable as the Global Price of Metal Index growth rate.

5.4. Baseline model with sign restriction identification

We implement sign restrictions, initially developed by Uhlig (2005), as an alternative identification method to the one used in the baseline model as well as in the three previous exercises (Robustness 1 to Robustness 3) that use short-term recursive identification. The imposed sign restrictions are motivated by economic theory and are restricted to the following: (i) a shock in NER has positive effects on IPI, PPI and CPI; (ii) a shock in IPI has positive effects on PPI and CPI; (iii) a PPI shock has a positive effect on CPI. Two additional restrictions are: a shock in GDP has a positive effect on DPM and a shock in DPM has a negative effect on IPI, PPI and CPI. These restrictions are evaluated at impact ($s = 0$) and after one quarter ($s = 1$). Given that the responses to the IPI and PPI shocks obtained by recursive identification comply with the required sign restrictions, both at the impact ($s = 0$) and after one quarter ($s = 1$), it is decided (for such shocks) perform sign constraint identification for the remaining shocks only. This approach, known as a mixture of sign restrictions and exclusion restrictions according to Kilian and Lütkepohl (2017), allows to considerably reduce the execution time of the algorithms by generating fewer rotation matrices in the identification process.

The identification by sign restrictions was carried out in each quarter of the time horizon of the sample, using a set of models made up of the one used in the baseline model and the three previous robustness exercises (TVP-VAR-SV, TVP-VAR-R2-SV, TVP-VAR-R3-SV, CVAR-SV and CVAR). We generate 5,000 rotations from a set of 10,000 IRFs paths obtained through recursive identification, in each period of the time horizon. For this purpose, an extension of the algorithm proposed by Gambetti and Musso (2017) was made, allowing the incorporation of exclusion restrictions and the use of parallel computing techniques.¹⁹

According to the Table 5 (column Robustness 4), based on the TVP-VAR-R3-SV model, we find that: (i) the ERPT into import prices, at the beginning of the sample, has a value of 0.43, exhibiting a fast growth with a steep slope until reaching a value of around 0.56 in 1999. Then, it exhibits a rapid reduction until 2009Q2, reaching a value of 0.47. During the 2009–2020 period, this ERPT shows a significant increase, reaching its maximum value around 0.57 at 2020Q4. Meanwhile, for the 2020–2022 period, a slight reduction is experienced, reaching a value close to 0.55 at the end of the sample; (ii) the ERPT into producer prices exhibit a similar behavior to import prices with a value of 0.22 at the beginning of the sample and reaching two local maximum values of 0.39 and 0.35 in the early stages of 1999 and 2020Q3 respectively; (iii) the ERPT into consumer prices has also a similar behavior show a value of 0.15 at the beginning of the sample, and two local peak values, the first occurring in early 1999 with 0.2 and the second in 2020Q2 with 0.15, oscillating with very little dispersion around 0.09 between 2003 and 2018.

Also for all models, Table 5 shows that the long-run ERPT are higher for import prices followed by producer prices and then by consumer prices, being this pattern the same as that observed in the baseline model and in the three previous robustness exercises, which is consistent with the PDC of McCarthy (2007). However, regarding the magnitude of the ERPTs, we observe that in the case of import prices and producer prices, the medians (column Robustness 4) are lower than those obtained by both the baseline model (see Table 3) as well as for the other robustness exercises (see Table 5, columns Robustness 1 to Robustness 3).

Regarding the evolution of the ERPTs throughout the sample, we observe: (i) in the case of import prices, each one of the ERPTs tends to a maximum value in 1999, followed by a fall until it rebounds in 2009; while, from the year 2020, all exhibit a negative slope, being more pronounced in some models than in others; (ii) in the case of the producer price, the ERPTs maintain a similar behavior as the case of import prices; (iii) in the case of the consumer price, the TVP-VAR-SV and TVP-VAR-R2-SV models start with a sharp fall in the ERPT, while the TVP-VAR-R3-SV model also does so but in a moderate way, exhibiting together with the ERPT of the CVAR-SV model an increasing trend until the end of 2020, to later adopt a negative slope until the beginning of 2022, where slope become positive; (iv) the median of the long-term ERPT is increasing for TVP-VAR-R3-SV and CVAR-SV models for all prices and decreasing for the TVP-VAR-SV and TVP-VAR-R2-SV models; (v) the TVP-VAR-R3-SV and CVAR-SV models show a higher ERPTs for all prices in the 2020Q1–2022Q4 sub-sample throughout the time horizon, followed by the ERPTs for the 2011Q1–2019Q4 sub-sample.

In summary, our findings are consistent with those of the baseline model, that is, we observe a growing trend in the ERPTs as well as a higher ERPT for the 2020Q1–2022Q4 subsample although the magnitudes are significantly lower. These smaller magnitudes, when compared to those obtained through Cholesky identification, can be attributed to the strict recursive identification enforced by

¹⁹ To carry out this stage, a PC equipped with an Intel Core i9-13900K processor (24 logical processors) and 64 GB DDR5 RAM at a speed of 6200Mhz was used. In addition, to optimize performance, the Parallel Computing Toolbox version 7.8 of MATLAB R2023b was used, using threads as fundamental units of execution.

this method. This approach is more restrictive than the pure sign restriction method and may consequently lead to spurious increases in responses at different horizons. For further details, please refer to Uhlig (2005), Mountford and Uhlig (2009), and Uhlig (2017). Finally, as part of our future research agenda, there is pending the development of a calculation strategy for the Log ML and the DIC using sign restrictions to establish a model ranking.

6. Conclusions

This paper uses a set of TVP-VAR-SV models based on Chan and Eisenstat (2018) to estimate and assess the evolution of the ERPT into three prices indexes for Peru over 1995Q2-2022Q4. The variables used, their ordering, and the shock identification scheme within the baseline VAR model are in line with the PDC model proposed by McCarthy (2007).

According to the two Bayesian selection criteria, the best-fitting model (TVP-VAR-R3-SV) allows most parameters to vary over time, emphasizing the role of SV. The estimations reveal that the ERPTs vary over time and are larger for import and producer prices than for consumer prices in line with the predictions of the PDC model. The short-term ERPT is larger than the long-term ERPT, a result consistent with the different degrees of price dollarization in import, producer and consumer prices indexes.

We find that the ERPTs into import and producer prices are high at the beginning of the sample; and begin to fall since 2000 (most notably from 2006, to a minimum in mid-2008) as a consequence of macroeconomic stabilization, an initial wave of de-dollarization policies, and surging commodity prices. However, unlike the existing literature for Peru (notably Winkelried (2014)), we find that the ERPTs tend to increase post-GFC due to higher price dollarization in import and producer prices.

Meanwhile, the ERPT into consumer prices exhibited relatively stable patterns, fluctuating between 10 and 20 percent throughout the entire sample period. Notably, it reached its lowest value during the 2000s, coinciding with a period of macroeconomic stabilization, adoption of IT regime, and the implementation of Law 28300. This finding supports the extensive evidence in favor of the low-inflation hypothesis proposed by Taylor (2000). In other words, economies with low and stable inflation, and monetary authorities that possess credibility in anchoring inflation expectations around the target, tend to experience a lower ERPT.

Interestingly, our main results are robust to using the strategy proposed by Lenza and Primiceri (2022) to correct extreme values associated to the impact of the Covid-19 crisis.

Distinguished from prior research, our study goes beyond merely quantifying ERPT; it delves into the nuanced analysis of short and long-term ERPT disparities and their intricate relationship with price dollarization along the supply chain. Moreover, we underscore the often-overlooked policy implications of these findings. Specifically, our research sheds light on how the combination of sticky dollar invoiced prices and exchange rate volatility can lead to inefficiencies in relative prices. This, in turn, can incentivize central banks in dollarized economies to intervene in the foreign exchange market, aiming to curb excessive exchange rate volatility and mitigate its detrimental effects on economic efficiency.

Furthermore, our findings draw attention to the recent surge in ERPT and its potential role in the rapid increase in inflation and imported prices witnessed in Peru in 2022. This highlights the need for proactive measures by the central bank to counteract second-round inflationary effects.

Finally, this paper does not address the drivers of the ERPT evolution. Therefore, a future research agenda could consider studying them. Additionally, it is necessary to examine the endogenous sources (i.e., the types of shocks) that drives the ERPT evolution using the methodologies proposed by Koop et al. (2009) and Leiva-León and Uzeda (2021). Some other issues remain as an agenda for further investigation: (i) explain in greater depth reasons why the long-term ERPT obtained by sign restriction are smaller in magnitude than those obtained with the baseline model (recursive identification); and (ii) obtain a calculation strategy for the Log ML and the DIC using sign restrictions in order to attain a ranking of the models under this approach.

CRedit authorship contribution statement

Gabriel Rodríguez: Formal analysis, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **Paul Castillo B.:** Conceptualization, Formal analysis, Methodology, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **Roberto Calero:** Conceptualization, Formal analysis, Methodology, Software, Writing – original draft, Writing – review & editing. **Rodrigo Salcedo Cisneros:** Conceptualization, Formal analysis, Methodology, Software, Writing – original draft, Writing – review & editing. **Miguel Ataurima Arellano:** Formal analysis, Methodology, Software, Writing – review & editing.

Data availability

Data will be made available on request.

References

- Albuquerque, C., Portugal, M., 2005. Pass-through from exchange rate to prices in Brazil: an analysis using time-varying parameters for the 1980-2002 period. *Rev. Econ.* 12 (1), 17–73.
- Aleem, A., Lahiani, A., 2014. A threshold vector autoregression model of exchange rate pass-through in Mexico. *Res. Int. Bus. Finance* 30 (C), 24–33. <https://doi.org/10.1016/j.ribaf.2013.05.001>.
- Alexius, A., Holmberg, M., 2017. Pass-through with low inflation and volatile exchange rates. *Research Papers in Economics* 2017:1. Department of Economics, Stockholm University.
- An, L., Wang, J., 2012. Exchange rate pass-through: evidence based on vector autoregression with sign restrictions. *Open Econ. Rev.* 23 (2), 359–380. <https://doi.org/10.1007/s11079-010-9195-8>.

- Aron, J., Macdonald, R., Muellbauer, J., 2014. Exchange rate pass-through in developing and emerging markets: a survey of conceptual, methodological and policy issues, and selected empirical findings. *J. Dev. Stud.* 50 (1), 101–143. <https://doi.org/10.1080/00220388.2013.847180>.
- Arratibel, O., Michaelis, H., 2014. The impact of monetary policy and exchange rate shocks in Poland: Evidence from a time-varying VAR, ECB Working Paper Series 1636.
- Bada, A., Olufemi, A., Tata, I., Peters, I., Bawa, S., Onwubiko, A., Onyowo, U., 2016. Exchange rate pass-through to inflation in Nigeria. *CBN J. Appl. Stat.* 7 (1), 49–70.
- Balcilar, M., Roubaud, D., Usman, O., Wohar, M., 2019. Testing the Asymmetric Effects of Exchange Rate and Oil Price Pass-Through in BRICS Countries: Does the state of the economy matter? Eastern Mediterranean University Discussion Paper Series 15–49.
- Baxa, J., Šestořád, T., 2019. The Czech exchange rate floor: Depreciation without inflation? Czech National Bank Working Paper 2019:1.
- Ben Ali, S., Tarek, J., 2014. Exchange rate pass-through in Tunisia: evidence from a time varying parameter model. *J. Dev. Areas* 48 (1), 145–164. <https://doi.org/10.1353/jda.2014.0007>.
- Bank for International Settlements, 2019. Monetary policy frameworks in EMEs: inflation targeting, the exchange rate and financial stability. BIS Annual Economic Report, Chapter II.
- Borensztein, E., Queijo, V., 2016. Exchange rate pass-through in South America: an overview. Inter-American Development Bank Working Paper Series 710.
- Buyandelger, O.E., 2015. Exchange rate pass-through effect and monetary policy in Mongolia: small open economy DSGE model. *Proc. Econ. Finance* 26, 1185–1192. [https://doi.org/10.1016/S2212-5671\(15\)00950-8](https://doi.org/10.1016/S2212-5671(15)00950-8).
- Calero, R., Rodríguez, G., Salcedo Cisneros, R., 2022. Evolution of the Exchange Rate Pass-Through into Prices in Peru: an Empirical Application Using TVP-VAR-SV Models. Working Paper 510. Department of Economics, Pontificia Universidad Católica el Perú.
- Castellares, R., 2017. Condiciones de mercado y calidad como determinantes del traspaso del tipo de cambio. *Rev. Est. Econ.* 33, 29–41.
- Castellares, R., Toma, H., 2020. Effects of a mandatory local currency pricing law on the exchange rate pass-through. *J. Int. Money Financ.* 106 (C), 102186. <https://doi.org/10.1016/j.jimonfin.2020.102186>.
- Castillo, P., Montoro, C., 2016. Income distribution and endogenous dollarization. In: Yamada, G., Winkelried, D. (Eds.), *Política y Estabilidad Monetaria en el Perú*. Universidad del Pacífico, Lima, pp. 237–268.
- Castillo, P., Montoro, C., Tuesta, V., 2007. Hechos estilizados de la economía peruana. *Rev. Est. Econ.* 14, 33–75.
- Castillo, P., Montoya, J., Quineche, R., 2016. From the “Great Inflation” to the “Great Moderation” in Peru: a time varying structural vector autoregressions analysis. Banco Central de Reserva del Perú Working Paper Series 2016-003.
- Çatık, A., Karaçuka, M., Gök, B., 2016. A time-varying parameter VAR investigation of the exchange rate pass-through in Turkey. *Panoeconomicus* 63 (5), 563–579. <https://doi.org/10.2298/PAN140307005C>.
- Ca’Zorzi, M., Hahn, E., Sánchez, M., 2007. Exchange rate pass-through in emerging markets. ECB Working Paper Series 739.
- Celeux, F., Forbes, F., Robert, C.P., Titterton, D.M., 2006. Deviance information criteria for missing data models. *Bayesian Anal.* 1 (4), 651–674. <https://doi.org/10.1214/06-BA122>.
- Chan, J.C.C., Eisenstat, E., 2015. Marginal likelihood estimation with the cross-entropy method. *Econom. Rev.* 34 (3), 256–285. <https://doi.org/10.1080/07474938.2014.944474>.
- Chan, J.C.C., Eisenstat, E., 2018. Bayesian model comparison for time-varying parameter VARs with stochastic volatility. *J. Appl. Econom.* 33 (4), 509–532. <https://doi.org/10.1002/jae.2617>.
- Chan, J.C.C., Grant, A., 2016. Fast computation of the deviance information criterion for latent variable models. *Comput. Stat. Data Anal.* 100, 847–859. <https://doi.org/10.1016/j.csda.2014.07.018>.
- Chan, J.C.C., Jeliazkov, I., 2009. Efficient simulation and integrated likelihood estimation in state space models. *Int. J. Math. Model. Numer. Optim.* 1 (1–2), 101–120. <https://doi.org/10.1504/IJMMNO.2009.030090>.
- Chávez, P.A., Rodríguez, G., 2023. Time changing effects of external shocks on macroeconomic fluctuations in Peru: empirical application using regime-switching VAR models with stochastic volatility. *Rev. World Econ.* 159, 505–544. <https://doi.org/10.1007/s10290-022-00474-1>.
- Chib, S., 1995. Marginal likelihood from the Gibbs output. *J. Am. Stat. Assoc.* 90, 1313–1321. <https://doi.org/10.1080/01621459.1995.10476635>.
- Chib, S., Jeliazkov, I., 2001. Marginal likelihood from the Metropolis-Hastings output. *J. Am. Stat. Assoc.* 96, 270–281. <https://doi.org/10.1198/016214501750332848>.
- Clark, T., Terry, S., 2010. Time variation in the inflation passthrough of energy prices. *J. Money Credit Bank.* 42 (7), 1419–1433. <https://doi.org/10.1111/j.1538-4616.2010.00347.x>.
- Cogley, T., Sargent, T., 2001. Evolving Post-World War II US inflation dynamics. *NBER Macroecon. Annu.* 16, 331–373. <https://doi.org/10.1086/654451>.
- Cogley, T., Sargent, T., 2005. Drifts and volatilities: monetary policies and outcomes in the post WWII US. *Rev. Econ. Dyn.* 8 (2), 262–302. <https://doi.org/10.1016/j.red.2004.10.009>.
- Comunale, M., Kunovac, D., 2017. Exchange rate pass-through in the euro area. European Central Bank Working Paper 2003.
- Contreras, A., Quispe, Z., Regalado, F., Martínez, M., 2017. Dolarización real en el Perú. *Rev. Est. Econ.* 33, 43–55.
- Corbo, V., Di Casola, P., 2018. Conditional exchange rate pass-through: Evidence from Sweden. Sveriges Riksbank Working Paper Series 352.
- Cornejo, G., Florian, D., Ledesma, A., 2022. La dinámica de la inflación doméstica ante cambios en cotizaciones internacionales de commodities, expectativas de inflación y tipo de cambio. Working Paper 2022-007. Central Reserve Bank of Peru.
- Cueva, R., 2018. Un análisis del traspaso del tipo de cambio: no linealidad y asimetría en México y Perú. *Rev. Est. Econ.* 35, 55–81.
- Dahem, A., Skander, S., Fatma, S.G., 2017. Time varying VAR analysis for disaggregated exchange rate pass-through in Tunisia. MPRA Paper 79759. University Library of Munich.
- Darvas, Z., 2001. Exchange rate pass-through and real exchange rate in EU candidate countries. Discussion Paper Series 10-01. Economic Research Centre of the Deutsche Bundesbank.
- De Bandt, O., Banerjee, A., Koźluk, T., 2008. Measuring long-run exchange rate pass-through. *Econ.: Open-Access, Open-Assess. E-J.* 2, 1–36. <https://doi.org/10.5018/economics-ejournal.ja.2008-6>.
- De Souza, R.M., Maciel, L.F.P., Pizzinga, A., 2013. State space models for the exchange rate pass-through: determinants and null/full pass-through hypotheses. *Appl. Econ.* 45 (36), 5062–5075. <https://doi.org/10.1080/00036846.2013.815397>.
- Donayre, L., Panovska, I., 2016. State-dependent exchange rate pass-through behavior. *J. Int. Money Financ.* 64 (C), 170–195. <https://doi.org/10.1016/j.jimonfin.2016.02.018>.
- Drenik, A., Perez, D.J., 2021. Domestic price dollarization in emerging economies. *J. Monet. Econ.* 122, 28–55. <https://doi.org/10.1016/j.jmoneco.2021.07.002>.
- Forbes, K., Hjortsoe, I., Nenova, T., 2018. The shocks matter: improving our estimates of exchange rate pass-through. *J. Int. Econ.* 114, 255–275. <https://doi.org/10.1016/j.jinteco.2018.07.005>.
- Frühwirth-Schnatter, S., Wagner, H., 2008. Marginal likelihoods for non-Gaussian models using auxiliary mixture sampling. *Comput. Stat. Data Anal.* 52 (10), 4608–4624. <https://doi.org/10.1016/j.csda.2008.03.028>.
- Gambetti, L., Musso, A., 2017. Loan supply shocks and the business cycle. *J. Appl. Econom.* 32 (4), 764–782.
- Gayaker, S., Ağaslan, E., Alkan, B., Çiçek, S., 2021. The deterioration in credibility, destabilization of exchange rate and the rise in exchange rate pass-through in Turkey. *Int. Rev. Econ. Finance* 76, 571–587. <https://doi.org/10.1016/j.iref.2021.07.004>.
- Gelfand, A.E., Dey, D.K., 1994. Bayesian model choice: asymptotics and exact calculations. *J. R. Stat. Soc. B* 56 (3), 501–514. <https://doi.org/10.1111/j.2517-6161.1994.tb01996.x>.

- Gondo, R., Pérez-Forero, F., 2019. Cross-border flows and the effect of global financial shocks in Latin America. Banco Central de Reserva del Perú Working Paper Series 2019-020.
- Hara, N., Hiraki, K., Ichise, Y., 2015. Changing exchange rate pass-through in Japan: Does it indicate changing pricing behavior? Bank of Japan Working Paper 15-E-4.
- Helmy, O., Fayed, M., Hussein, K., 2018. Exchange rate pass-through to inflation in Egypt: a structural VAR approach. *Rev. Econ. Polit. Sci.* 3 (2), 2–19. <https://doi.org/10.1108/REPS-07-2018-001>.
- Hernández, K., Leblebicioğlu, A., 2012. A regime-switching analysis of pass-through. *Rev. World Econ.* 148 (3), 523–552. <https://doi.org/10.1007/s10290-012-0120-7>.
- Jašová, M., Moessner, R., Takáts, E., 2019. Exchange rate pass-through: what has changed since the crisis? *Int. J. Cent. Bank.* 15 (3), 27–58.
- Jiménez-Rodríguez, R., Morales-Zumaquero, A., 2016. A new look at exchange rate pass-through in the G-7 countries. *J. Policy Model.* 38 (5), 985–1000. <https://doi.org/10.1016/j.jpolmod.2016.06.007>.
- Jooste, C., Jhaveri, Y., 2014. The determinants of time-varying exchange rate pass-through in South Africa. *South Afr. J. Econ.* 82 (4), 603–615. <https://doi.org/10.1111/saje.12058>.
- Karagöz, M., Demirel, B., Bozdağ, E.G., 2016. Pass-through effect from exchange rates to the prices in the framework of inflation targeting policy: a comparison of Asia-Pacific, South American and Turkish economies. *Proc. Econ. Finance* 38, 438–445. [https://doi.org/10.1016/S2212-5671\(16\)30215-5](https://doi.org/10.1016/S2212-5671(16)30215-5).
- Khemiri, R., Ben Ali, M.S., 2013. Exchange rate pass-through and inflation dynamics in Tunisia: a Markov-switching approach. *Econ.: Open-Access, Open-Assess. E-J.* 7, 1–30. <https://doi.org/10.5018/economics-ejournal.ja.2013-43>.
- Kilian, L., Lütkepohl, H., 2017. *Structural Vector Autoregressive Analysis*. Cambridge University Press, Cambridge.
- Kim, Y., 1990. Exchange rates and import prices in the United States: a varying-parameter estimation of exchange-rate pass-through. *J. Bus. Econ. Stat.* 8 (3), 305–315. <https://doi.org/10.2307/1391967>.
- Kim, S., Shephard, N., Chib, S., 1998. Stochastic volatility: likelihood inference and comparison with ARCH models. *Rev. Econ. Stud.* 65 (3), 361–393.
- Koop, G., León-González, R., Strachan, R.W., 2009. On the evolution of the monetary policy transmission mechanism. *J. Econ. Dyn. Control* 33, 997–1017. <https://doi.org/10.1016/j.jedc.2008.11.003>.
- Leiva-León, D., Uzeda, L., 2021. Endogenous time variation in vector autoregressions. *Rev. Econ. Stat.* 1 (45). https://doi.org/10.1162/rest_a_01038.
- Lenza, M., Primiceri, G., 2022. How to estimate a vector autoregression after March 2020. *J. Appl. Econom.* 37 (4), 688–699. <https://doi.org/10.1002/jae.2895>.
- Liu, H., Chen, X., 2017. The imported price, inflation and exchange rate pass-through in China. *Cogent Econ. Finance* 5 (1), 1–13. <https://doi.org/10.1080/23322039.2017.1279814>.
- López-Villavicencio, A., Mignon, V., 2016. Exchange rate pass-through in emerging countries: do the inflation environment, monetary policy regime and central bank behavior matter? *J. Int. Money Financ.* 79, 20–38. <https://doi.org/10.1016/j.jimonfin.2017.09.004>.
- Maertens Odría, L., Castillo, P., Rodríguez, G., 2012. Does the exchange rate pass-through into prices change when inflation targeting is adopted? The Peruvian case study between 1994 and 2007. *J. Macroecon.* 34 (4), 1154–1166. <https://doi.org/10.1016/j.jmacro.2012.07.001>.
- Marodin, F., Portugal, M., 2019. Exchange rate pass-through in Brazil: a Markov switching DSGE estimation for the inflation targeting period. *Russ. J. Money Finance* 78 (1), 36–66. <https://doi.org/10.31477/rjmf.201901.36>.
- McCarthy, J., 2007. Pass-through of exchange rates and import prices to domestic inflation in some industrialized economies. *East. Econ. J.* 33 (4), 511–537. <https://doi.org/10.2139/ssrn.249576>.
- Miller, S., 2003. Estimación del pass-through del tipo de cambio a precios: 1995-2002. *Rev. Est. Econ.* 10, 137–170.
- Montoro, C., 2006. Dolarización de precios. Nota de Estudio 2006-14. Banco Central de Reserva del Perú.
- Mountford, A., Uhlig, H., 2009. What are the effects of fiscal policy shocks?. *J. Appl. Econ.* 24, 960–992. <https://doi.org/10.1002/jae.1079>.
- Moussa, Z., 2016. How big is the comeback? Japanese exchange rate pass-through assessed by time-varying FAVAR. LEMNA Working Paper. Université de Nantes.
- Nakajima, J., 2011. Time-varying parameter VAR model with stochastic volatility: an overview of methodology and empirical applications. *Monet. Econ. Stud.* 29, 107–142.
- Ozkan, I., Erden, L., 2015. Time-varying nature and macroeconomic determinants of exchange rate pass-through. *Int. Rev. Econ. Finance* 38, 56–66. <https://doi.org/10.1016/j.iref.2015.01.007>.
- Palleja, M., 2019. Conditional Exchange Rate Pass-Through: A DSGE Model Approach. 2018 Central Bank Award Rodrigo Gómez. Center for Latin American Monetary Studies.
- Pan, S., 2018. The impact of RMB exchange rate fluctuation on price level in China: an empirical analysis based on the vector error correction model. *Int. J. Econ. Finance* 10 (5), 184–196. <https://doi.org/10.5539/ijef.v10n5p184>.
- Patra, M.D., Kumar, J., John, J., 2018. Non-linear, asymmetric and time-varying exchange rate pass-through: Recent evidence from India. Reserve Bank of India Working Paper Series 02.
- Pérez-Forero, F., Vega, M., 2015. Asymmetric exchange rate pass-through: Evidence from Peru. Banco Central de Reserva del Perú Working Paper Series 2015-011.
- Ponomarev, I., Trunin, P., Ulyukhaev, A., 2016. Exchange rate pass-through in Russia. *Probl. Econ. Trans.* 58 (1), 54–72. <https://doi.org/10.2139/ssrn.2458881>.
- Primiceri, G.E., 2005. Time varying structural vector autoregressions and monetary policy. *Rev. Econ. Stud.* 72 (3), 821–852. <https://doi.org/10.1111/j.1467-937X.2005.00353.x>.
- Razafindrabé, T., 2016. A multi-country DSGE model with incomplete exchange rate pass-through: an application for the Euro-area. *Econ. Model.* 52, 78–100. <https://doi.org/10.1016/j.econmod.2015.03.003>.
- Rincón-Castro, H., Rodríguez-Niño, N., 2018. Nonlinear state and shock dependence of exchange rate pass-through on prices. Bank for International Settlements Working Paper 690.
- Rodríguez, G., Villanueva Vega, P., Castillo, P., 2018. Driving economic fluctuations in Peru: the role of the terms of trade. *Empir. Econ.* 55 (3), 1089–1119. <https://doi.org/10.1007/s00181-017-1318-2>.
- Rodríguez, G., Vassallo, R., Castillo B., P., 2023. Effects of external shocks on macroeconomic fluctuations in Pacific Alliance countries. *Econ. Model.* 124, 106302. <https://doi.org/10.1016/j.econmod.2023.106302>.
- Rodríguez, G., Castillo B., P., Ojeda Cunya, J.A., 2024. Time-varying effects of external shocks on macroeconomic fluctuations in Peru: empirical application using TVP-VAR-SV models. *Open Econ. Rev.* <https://doi.org/10.1007/s11079-023-09742-5>.
- Rossini, R., Vega, M., Quispe, Z., Pérez-Forero, F., 2016. Expectativas de inflación y dolarización en el Perú. *Rev. Est. Econ.* 31, 71–84.
- Sansone, A., Justel, S., 2016. Exchange rate pass-through to prices: VAR evidence for Chile. *J. Econ. Chil. (Chil. Econ.)* 19 (1), 20–37.
- Sasaki, Y., 2019. Pass-through effect: Exchange market impact on prices - Is the pass-through effect on Japanese imports declining? *Financ. Rev.* 27, 118–143.
- Sasaki, Y., Yoshida, Y., 2018. Decomposition of Japan's trade balance. *Int. Rev. Econ. Finance* 56 (1), 507–537. <https://doi.org/10.1016/j.iref.2017.12.007>.
- Sasaki, Y., Yoshida, Y., Kansho Otsubo, P., 2022. Exchange rate pass-through to Japanese prices: Import prices, producer prices, and the core CPI. *J. Int. Money Financ.* 123, 102599. <https://doi.org/10.1016/j.jimonfin.2022.102599>.
- Sekine, T., 2006. Time-varying exchange rate pass-through: Experiences of some industrial countries. Bank for International Settlements Working Paper 202.
- Shambaugh, J., 2008. A new look at pass-through. *J. Int. Money Financ.* 27, 560–591. <https://doi.org/10.1016/j.jimonfin.2008.01.009>.
- Shioji, E., 2012. The evolution of the exchange rate pass-through in Japan: a re-evaluation based on time-varying parameter VARs. *Publ. Policy Rev.* 8 (1), 67–92.
- Shioji, E., 2014. A pass-through revival. *Asian Econ. Policy Rev.* 9, 120–138. <https://doi.org/10.1111/aepr.12053>.
- Shioji, E., 2015. Time varying pass-through: will the yen depreciation help Japan hit the inflation target? *J. Jpn. Int. Econ.* 37, 43–58. <https://doi.org/10.1016/j.jjie.2015.06.001>.
- Spiegelhalter, D.J., Carlin, B.P., Van der Linde, A., 2002. Bayesian measures of model complexity and fit. *J. R. Stat. Soc. Ser. B* 64 (4), 583–639. <https://doi.org/10.1111/1467-9868.00353>.

- Takhtamanova, Y., 2010. Understanding changes in exchange rate pass-through. *J. Macroecon.* 32 (4), 1118–1130. <https://doi.org/10.1016/j.jmacro.2010.04.004>.
- Taylor, J.B., 2000. Low inflation, pass-through, and the pricing power of firms. *Eur. Econ. Rev.* 44 (7), 1389–1408. [https://doi.org/10.1016/S0014-2921\(00\)00037-4](https://doi.org/10.1016/S0014-2921(00)00037-4).
- Tunç, C., 2017. A survey on exchange rate pass through in emerging markets. *Bull. Econ. Theory Anal.* 2 (3), 205–233. <https://doi.org/10.25229/beta.334253>.
- Tunç, C., Kılıç, M., 2018. Exchange rate pass-through in a small open economy: a structural VAR approach. *Bull. Econ. Res.* 70 (4), 410–422. <https://doi.org/10.1111/boer.12162>.
- Uhlig, H., 2005. What are the effects of monetary policy on output? Results from an agnostic identification procedure. *J. Monet. Econ.* 52 (2), 381–419.
- Uhlig, H., 2017. Shocks, sign restrictions, and identification. In: Honoré, B., Pakes, A., Piazzesi, M., Samuelson, L. (Eds.), *Advances in Economics and Econometrics: Eleventh World Congress*. In: *Econometric Society Monographs*. Cambridge University Press, Cambridge, pp. 95–127.
- Winkelried, D., 2003. ¿Es asimétrico el pass-through en el Perú?: un análisis agregado. *Rev. Est. Econ.* 10, 171–199.
- Winkelried, D., 2014. Exchange rate pass-through and inflation targeting in Peru. *Empir. Econ.* 46 (4), 1181–1196. <https://doi.org/10.1007/s00181-013-0715-4>.
- Wu, J.C., Xia, F.D., 2016. Measuring the macroeconomic impact of monetary policy at the zero lower bound. *J. Money Credit Bank.* 48 (2–3), 253–291. <https://doi.org/10.1111/jmcb.12300>.